SECURE MICROFINANCE AND WELFARE MANAGEMENT PLATFORM USING BLOCKCHAIN AND AI

INDIVIDUAL THESIS

THARUSHIKA A J M H

IT22194626

Individual component: Identity Verification and Configurable Welfare Eligibility Assessment

Group ID: R26-IT-017

Report submitted in partial fulfilment of the requirements for the

BSc Information Technology

Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

October 2026

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ABSTRACT

This research presents an individual component of SmartGrama, a digital governance platform that combines digital identity, welfare assessment and permissioned blockchain technologies. The component addresses the need for a welfare eligibility process that is structured, explainable, privacy-aware and auditable. The proposed solution connects verified identity and KYC information with an Aswesuma-oriented welfare application, a configurable eligibility assessment engine, authorized officer review and blockchain-based audit recording. Six welfare dimensions are considered: education, health and special needs, economic level and income, assets and ownership, housing conditions, and family demography. The eligibility engine produces a prototype score and welfare category using configurable weights and thresholds. The score is advisory and does not replace the official decision-maker; an authorized officer makes the final APPROVED or REJECTED decision. The final decision can then be represented as a minimized blockchain audit payload containing references such as the application ID, DID reference, rule version, result, timestamp and integrity hash. SHA-256 hashing is used to provide tamper-evidence for the audit data. The component was developed using a modular service architecture and a blockchain provider interface intended for Hyperledger chain code integration. Controlled automated testing reported 64/64 passing tests, including 17/17 blockchain-audit tests, with zero TypeScript compilation errors. The research does not claim to reproduce the official production Aswesuma PMT algorithm; instead, it contributes a configurable research prototype that can accept validated policy parameters in future deployments. The main contribution is the integration of privacy-aware identity context, explainable welfare assessment, human decision-making and auditable permissioned blockchain recording within one component architecture.

Keywords: decentralized identity, welfare eligibility, Aswesuma, Hyperledger Fabric, smart contracts

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to my supervisor and academic staff for their guidance, feedback and encouragement throughout this research. I also acknowledge the support of my group members who worked on the blockchain, artificial intelligence and natural language processing components of the SmartGrama platform. Their collaboration helped define the integration boundaries of my individual component. I am grateful to the researchers, standards organizations and public institutions who's technical and policy resources informed the design of the decentralized identity, welfare assessment, privacy and blockchain aspects of this work. Finally, I thank my family and friends for their continued encouragement and support throughout the completion of this thesis.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
API	Application Programming Interface
DID	Decentralized Identifier
KYC	Know Your Customer
PII	Personally Identifiable Information
PMT	Proxy Means Test
VC	Verifiable Credential
WBB	Welfare Benefits Board
DLT	Distributed Ledger Technology
SHA-256	Secure Hash Algorithm 256-bit
UI	User Interface
UX	User Experience

1 INTRODUCTION

1.1 Overall Project Context

SmartGrama is a group research and development project intended to improve access to welfare and microfinance-related services through a digital platform. The project explores how digital identity, artificial intelligence (AI), natural language processing (NLP), and permissioned blockchain technology can support different stages of service delivery. Its overall motivation is to investigate practical ways of improving information access, reducing fragmented record handling, supporting administrative workflows, and strengthening the reliability of records used by participating services.

Welfare and microfinance services may involve sensitive personal information, household circumstances, financial details, and decisions that affect citizens' access to assistance. Consequently, a digital platform in this domain must consider not only functional requirements but also privacy, security, explainability, accountability, and appropriate human oversight.

SmartGrama is organized into individual research components that address different technical problems within the wider platform. The component presented in this thesis focuses on digital identity context, welfare eligibility assessment, officer decision-making, and blockchain-supported auditability. It is designed as a distinct component that can contribute to the overall SmartGrama system while maintaining a clearly defined individual research scope.

The work reported in this thesis evaluates the design and prototype implementation of that component. It does not claim that every SmartGrama component has been integrated into a complete production system.

1.2 Individual Component Context

The individual component investigated in this thesis is a privacy-aware welfare eligibility assessment and blockchain audit framework supported by verified digital identity. It focuses on an Aswesuma-oriented application workflow in which identity-related information supports the application process, household information is

assessed through configurable rules, and an authorized officer retains responsibility for the final decision.

The proposed workflow contains the following main stages:

1. Identity and KYC context: establish the identity-verification context required by the application workflow.
2. Welfare application: capture the information needed to assess an applicant's household circumstances.
3. Eligibility assessment: process the available information using a configurable, rule-based assessment engine.
4. Assessment explanation: produce a prototype score or category that helps explain the assessment outcome.
5. Officer review: allow an authorized officer to review the assessment and make the final approval or rejection decision.
6. Audit recording: prepare an audit record representing the authorized decision and the relevant integrity metadata.
7. Audit verification: retrieve and verify the stored prototype audit record and its associated hash.

The eligibility assessment considers six dimensions: education; health and special needs; economic level and income; assets and ownership; housing conditions; and family demography. These dimensions provide the structure for the prototype's configurable assessment process. The coefficients, weights, and thresholds are research-prototype parameters; they must not be interpreted as a reproduction of the official production Aswesuma Proxy Means Test (PMT) algorithm.

The assessment score is intended to support administrative review rather than replace it. An automated result alone does not establish the official entitlement of an applicant. The authorized officer remains responsible for the final decision, including consideration of cases that may require clarification or further review.

The blockchain aspect addresses the need to maintain a traceable history of relevant decisions. In the completed individual prototype, a backend service and mock blockchain provider support audit-record operations and hash verification. A provider

interface was designed to support the intended blockchain integration boundary. However, actual Hyperledger Fabric network and chain code integration was left for integration by the wider team. Therefore, the evaluation in this thesis must distinguish the tests performed against the mock provider from any later tests performed against a real Fabric network.

1.3 Problem Statement

Welfare eligibility assessment requires consideration of several aspects of a household's circumstances, potentially including income, assets, housing conditions, health-related needs, education, and family composition. Handling this information requires a process that supports structured assessment while protecting the privacy of applicants.

Research on proxy means testing and welfare targeting in Sri Lanka demonstrates that the design of eligibility assessment is an important policy and technical issue [1], [2]. However, an eligibility process implemented as a digital application must address more than the calculation of an assessment result. It must also support understandable outcomes, authorized decision-making, appropriate information handling, and a reliable record of administrative actions.

Several related challenges motivate this research. First, a configurable assessment mechanism is needed to process multiple household dimensions in a consistent way without presenting an unvalidated prototype formula as an official government algorithm. Second, the final decision must remain under appropriate human authority rather than being determined solely by an automated score. Third, the system must distinguish sensitive personal information required for assessment from the smaller amount of information needed for audit and integrity verification. Finally, the prototype requires a way to record and verify audit information without claiming that a mock implementation provides the same operational properties as a deployed permissioned blockchain.

The central research problem is therefore:

How can a privacy-aware welfare eligibility component combine verified digital identity context, configurable and explainable household assessment, authorized

human decision-making, and verifiable audit records while minimizing unnecessary exposure of sensitive information?

This problem is addressed through the design and implementation of a modular prototype. The prototype includes a configurable assessment workflow and backend audit services that use a mock blockchain provider. Its hash-verification functionality is evaluated through controlled tests. Integration with a live Hyperledger Fabric network remains outside the completed individual implementation and is identified as a limitation and future integration activity.

1.4 Research gap

Existing work provides foundations for the individual areas relevant to this research. Research on welfare targeting and proxy means testing provides context for the assessment of household circumstances [1], [2]. Decentralized identity specifications provide a model for representing and resolving digital identifiers [3]. Hyperledger Fabric documentation describes a permissioned blockchain platform and its transaction-processing architecture [4]. These sources provide useful foundations, but the existence of individual technologies does not, by itself, establish that they have been successfully combined in a particular welfare workflow.

The research gap addressed by this component is the need to bring together the following elements within a clearly bound and testable prototype:

- identity-verification context linked to a welfare application;
- configurable assessment across multiple household dimensions;
- an explainable prototype result that supports, rather than replaces, an authorized officer's decision;
- separation of sensitive application information from the audit information required for integrity checking; and
- a modular audit-record service with hash verification and a defined integration boundary for a permissioned blockchain.

A further gap concerns the distinction between designing for blockchain integration and demonstrating a live blockchain implementation. A prototype can test its audit-record structure, service behavior, and hash-verification logic using a mock provider.

However, these tests alone cannot demonstrate transaction endorsement, ordering, consensus-related behavior, access-control enforcement, or ledger persistence on an operational Hyperledger Fabric network. This thesis therefore treats the mock-provider results as evidence of prototype behavior and does not use them as proof of completed Fabric integration.

The contribution is consequently an integrated component design and tested backend prototype, rather than a claim to have created a new official welfare eligibility algorithm or completed a production blockchain deployment.

1.5 Aim and objectives

The aim of this research is to design, implement, and evaluate a privacy-aware welfare eligibility assessment component supported by verified digital identity context, configurable assessment rules, authorized officer review, and a hash-verifiable audit mechanism designed for permissioned blockchain integration.

The research is guided by the following objectives.

Objective 1 - Design the application workflow. Define the relationship between identity and KYC context, welfare application information, eligibility assessment, officer review, and audit recording.

Objective 2 - Develop configurable eligibility assessment. Implement a rule-based prototype that considers six household dimensions: education, health and special needs, economic level and income, assets and ownership, housing conditions, and family demography.

Objective 3 - Provide an explainable prototype result. Generate an assessment score or category using configurable prototype parameters, while clearly distinguishing the result from an official government eligibility determination.

Objective 4 - Preserve human decision-making. Represent an authorized officer's final approval or rejection decision separately from the automatically calculated assessment result.

Objective 5 - Implement privacy-aware audit functionality. Develop backend audit-record operations, a mock blockchain provider, and SHA-256 hash-based integrity verification, with a defined service boundary for future Hyperledger Fabric integration.

Objective 6 - Evaluate prototype correctness and limitations. Assess the implemented functionality through controlled software tests, including audit-record operations, hash verification, and relevant workflow behavior. Identify which requirements have been tested in the individual prototype and which depend on later team integration.

These objectives address the technical implementation and evaluation aspects of the research. They also provide a basis for assessing whether the prototype responds to the problem identified in Section 1.3.

1.6 Research questions or hypotheses

Four research questions guide the investigation.

RQ1: Can a configurable welfare eligibility engine provide a consistent and explainable prototype assessment using multiple household dimensions?

RQ2: Can sensitive welfare and identity information be separated from the information required for an audit record and integrity verification?

RQ3: Can a human-in-the-loop workflow preserve authorized administrative decision-making while benefiting from automated eligibility calculation?

RQ4: Can a modular audit service provide testable hash-verification and audit-record functionality while defining a practical integration boundary for a permissioned blockchain?

These questions are aligned with the implemented scope. RQ1 concerns the assessment engine's behavior, not its equivalence to an official PMT model. RQ2 concerns the design and handling of audit payloads and must be evaluated against the actual implementation. RQ3 concerns the separation between automated assessment and the authorized final decision. RQ4 concerns the implemented mock-provider service and hash verification; it does not imply that live Fabric transaction behavior has already been demonstrated.

1.7 Scope limitations and assumptions

The research covers the individual welfare eligibility and audit component of SmartGrama. Its scope includes:

- The design of the identity and KYC context for a welfare application;
- The representation of an Aswesuma-oriented application workflow;
- Configurable, rule-based assessment using six household dimensions;
- Production of a prototype assessment score or category;
- Separation of the assessment result from the authorized officer's final decision;
- Backend audit-record operations using a mock blockchain provider;
- SHA-256 hash-based integrity verification;
- Controlled software testing of the implemented prototype; and
- Definition of the intended integration boundary for Hyperledger Fabric and chain code.

The component is designed to work within the wider SmartGrama project. The thesis concentrates on the individual's research contribution rather than claiming ownership of all group components.

The following limitations define how the results should be interpreted.

Prototype eligibility parameters: The assessment coefficients and thresholds are configurable research parameters. The prototype does not claim to reproduce the official production Aswesuma PMT algorithm or to establish actual government eligibility.

Mock blockchain provider: Audit-record operations and hash verification were tested using the mock provider. Actual Hyperledger Fabric network and chain code integration was left for team integration. Consequently, the individual prototype results do not establish the performance or security of a live Fabric deployment.

Controlled testing: Automated tests can provide evidence that implemented functions behave as expected for the tested inputs and conditions. They do not, by themselves, prove suitability for nationwide deployment or effectiveness with all real-world welfare cases.

Privacy and security validation: Hashing supports integrity checking but does not encrypt the underlying information or guarantee confidentiality. Any claim about which fields are stored off-chain or included in an audit payload must be verified against the final implementation and its actual data structures.

External validation: No claim of formal validation by government authorities, welfare beneficiaries, or other external stakeholders is made unless supported by documented evidence.

The design assumes that:

1. Identity-related information and application information are supplied through authorized application workflows;
2. Eligibility rules and thresholds can be configured and reviewed independently of the software implementation;
3. The final welfare decision is made by an authorized officer;
4. Audit records should contain only the information necessary for their defined verification and traceability purposes;
5. The mock provider can be used to test the backend's expected audit-service behavior without being treated as a real blockchain; and
6. Future integration with Hyperledger Fabric will require separate configuration, access-control decisions, chain code integration, and end-to-end verification.

These assumptions guide the prototype but are not substitutes for validation in a real deployment.

1.8 Individual contribution

The individual contribution of this research is the design and implementation of a configurable welfare eligibility and audit component within SmartGrama. The contribution focuses on linking structured household assessment with authorized human review and a backend audit mechanism while recognizing the privacy requirements associated with welfare data.

The principal individual contributions are:

1. Component workflow design: defining the relationship between identity context, welfare application, assessment, officer decision, and audit operations.
2. Configurable assessment approach: implementing a prototype assessment mechanism based on six household dimensions, with configurable parameters rather than a claim to reproduce an official production algorithm.
3. Human decision boundary: separating the calculated assessment result from the officer's final approval or rejection decision.

4. Backend audit functionality: implementing the audit-record service and mock blockchain provider used by the individual prototype.
5. Integrity verification: implementing and testing SHA-256-based hash verification for the supported audit operations.
6. Integration design: defining the provider boundary through which the backend is intended to connect to Hyperledger Fabric and chain code.
7. Prototype evaluation: using controlled software tests to assess implemented behavior and document the distinction between mock-provider testing and live blockchain integration.

1.9 Thesis structure

Chapter 1: Introduction establishes the project context, individual component, problem statement, research gap, aim, objectives, research questions, scope, limitations, assumptions, and individual contribution.

Chapter 2: Literature Review and Related Work examines the relevant literature and technical foundations for welfare eligibility assessment, decentralized identity, permissioned blockchain systems, integrity verification, and privacy-aware record management. It synthesizes the evidence to establish the research gap.

Chapter 3: Research Methodology describes the research design, study materials, development and testing procedures, evaluation measures, data analysis approach, and ethical, privacy, security, and reproducibility considerations.

Chapter 4: Product and Solution Design specifies the component requirements, architecture, design alternatives, detailed service and data design, and relevant professional practices.

Chapter 5: Implementation, Integration and Testing documents the development environment, assessment engine, backend audit service, mock provider, hash-verification implementation, testing approach, and integration boundary. It distinguishes implemented functionality from the live Fabric integration left for team integration.

Chapter 6: Results and Validation presents the evidence obtained from controlled testing and other documented evaluation activities. It reports technical results within the limits of the evidence and does not present unperformed live-network or stakeholder tests as completed.

Chapter 7: Discussion interprets the findings, compares the contribution with related work, evaluates the research and product contributions, and discusses limitations, threats to validity, and implications.

Chapter 8: Commercialization and Sustainability considers the component's potential value, target users, competitive position, sustainability, scalability, compliance requirements, and possible contribution to relevant Sustainable Development Goals.

Chapter 9: Conclusion and Future Directions summarizes the research, evaluates objective achievement, states the individual contributions, and identifies future work, including further integration and testing with Hyperledger Fabric.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

The purpose of this literature review is to examine the technologies, standards, research findings, and existing approaches relevant to privacy-preserving decentralized identity verification and smart-contract-based welfare eligibility assessment. The review establishes the technical and conceptual foundation for the SmartGrama research component and identifies the research gap addressed through the design and development of the proposed solution.

The review focuses on four main areas: decentralized digital identity, biometric and document-based identity verification, blockchain-supported welfare services, and smart-contract-based eligibility assessment. Privacy protection, traceability, and human oversight are considered across these areas because they influence the suitability of the proposed solution for public welfare services.

2.2 Domain and theoretical foundations

Decentralized identity: The W3C DID Core specification defines decentralized identifiers as identifiers designed for verifiable decentralized digital identity [1]. In SmartGrama, a DID reference provides an identity-layer identifier without requiring the complete citizen profile to be written to the blockchain.

Verifiable credentials: The W3C Verifiable Credentials Data Model provides a standardized model for expressing verifiable claims [2]. This supports the architectural principle that identity evidence can be represented through credentials or references instead of repeatedly exposing raw identity documents.

Welfare targeting and proxy means testing: Research by the World Bank examined proxy means testing for targeting welfare benefits in Sri Lanka [4], [5]. The general principle is relevant because household characteristics can be used to estimate welfare status when direct measurement of household welfare is difficult. This research uses multidimensional concepts but does not claim to reproduce the World Bank model or the official current Aswesuma production coefficients.

Blockchain and smart contracts: Hyperledger Fabric is a permissioned blockchain platform suitable for applications where participating organizations are known and governed [3]. This is appropriate for the proposed welfare audit context because unrestricted public disclosure is not desirable for sensitive welfare information.

Integrity hashing: SHA-256 produces a cryptographic digest that can be used to detect changes in data [9]. The component uses hashing for tamper-evidence. It is not used as encryption and does not by itself provide confidentiality.

Biometric verification: NIST research on face recognition has identified demographic differences among evaluated algorithms [10]. Therefore, biometric verification is treated as one identity-proofing mechanism requiring careful evaluation rather than as an infallible identity decision.

2.3 Related research

The reviewed research provides the theoretical and technical foundations for the SmartGrama component by examining decentralized identity standards, facial recognition, welfare targeting, blockchain technology, and cryptographic integrity. The W3C Decentralized Identifiers (DID) specification and Verifiable Credentials Data Model provide standards for identity representation and the exchange of claims, while NIST research highlights the importance of evaluating facial recognition performance across demographic groups. Studies on welfare targeting in Sri Lanka emphasize the need for an appropriate, programmed-specific eligibility methodology, and Hyperledger Fabric provides a permissioned blockchain framework for executing smart contracts and recording auditable transactions. Based on these foundations, SmartGrama connects citizen identity verification using facial and document evidence, predefined welfare eligibility rules, authorized officer review, and blockchain-based auditability. However, the effectiveness of this approach depends on the reliability of the verification mechanisms, the suitability of the eligibility rules, the quality of input data, and the correctness of the implementation. Therefore, the component does not claim to establish national welfare policy or automatically guarantee improved processing speed, fraud prevention, fairness, or citizen trust; these outcomes must be demonstrated through appropriate implementation and evaluation.

2.4 Existing systems products and techniques

Centralized identity systems typically rely on an organization or service that manages identity records and verification. This approach provides clear administrative responsibility and a central point for managing identity information; however, it can create dependencies on the availability and security of the central service. Decentralized identifiers offer an alternative model in which identifiers can be designed to operate independently of a single centralized registry [1]. Although this approach may support interoperability and more flexible identity management, it does not eliminate the need for trusted identity evidence, secure applications, and clear governance. Therefore, the relevant design consideration for SmartGrama is not whether decentralized identity is universally superior, but whether identity services, verification evidence, and access controls can support the intended welfare workflow while maintaining appropriate privacy and accountability.

Identity verification can be performed using documents, facial recognition, existing digital identities, authorized officer review, or a combination of these techniques. Document verification provides documentary evidence associated with a claimed identity, although documents may be invalid, inconsistent, or difficult to validate. Facial verification supports comparison between a captured face and a registered reference, but its performance depends on the algorithm, image quality, and operating conditions. NIST's research highlights the importance of considering variations in facial recognition performance [3]. An existing digital identity provides a starting point for a citizen's workflow, but the existence of an identity does not independently prove who is presenting it. Authorized officer review allows evidence and exceptional cases to be examined, although clear procedures and accountable human judgment are required. A combined verification workflow brings identity information, supporting evidence, and human review into one process, but its effectiveness depends on reliable integration and appropriate evaluation.

Table 1 : Comparison of identity-verification techniques

Technique	**Potential benefit**	**Main limitation**	**Relevance to SmartGrama**

Identity-document verification	Provided documentary evidence associated with a claimed identity	Documents may be invalid, inconsistent, or difficult to validate	NIC or other document evidence
Facial verification	Supports comparison between a captured face and a registered reference	Performance depends on the algorithm, image quality, and operating conditions	Biometric identity evidence
Existing digital identity	Provides a starting point for the citizen's workflow	The existence of an identity does not, by itself, prove who is presenting it	Existing SmartGrama identity
Authorized officer review	Supports examination of evidence and exceptional cases	Requires clear procedures and accountable human judgment	Verification review and decision
Combined verification workflow	Brings identity information, evidence, and human review into one process	Requires reliable integration and evaluation	Proposed component workflow

A combined verification workflow provides a structured way to handle different forms of identity evidence. However, its effectiveness must be established through appropriate testing. The comparison identifies the design rationale for combining evidence and review; it does not demonstrate that the implemented component has experimentally outperformed biometric-only or document-only systems.

Manual eligibility assessment allows officers to interpret evidence and address exceptional cases. However, repeatedly applying eligibility criteria manually may be time-consuming and can lead to inconsistent outcomes when procedures are unclear. Software-based rule assessment represents eligibility conditions explicitly so that they can be evaluated against application data. When the same valid inputs are processed using the same correctly implemented rules, the assessment should produce consistent results. Smart-contract-based assessment applies business logic through a blockchain framework, and Hyperledger Fabric supports smart contracts that generate transactions recorded on the ledger [6]. This provides an opportunity to connect rule execution with a traceable transaction history. Nevertheless, manual review, conventional software-based assessment, and smart-contract execution serve different purposes. Manual review supports human judgment, conventional software executes application-level rules, and smart contracts execute business logic within a blockchain environment. Their suitability depends on governance, integration requirements, data quality, security, and audit needs. For SmartGrama, the proposed design connects predefined eligibility-rule execution with an officer-led final decision, allowing the consistency of automated assessment to be considered without assuming that automation should replace administrative judgment.

Conventional databases support the storage, searching, updating, and retrieval of application information, whereas blockchain ledgers provide a transaction history governed by the rules of a participating network. These approaches can complement one another rather than being treated as mutually exclusive. In welfare applications, detailed personal and biometric information should be handled through appropriately controlled services, while the audit trail should contain only the information necessary to support traceability and verification. Depending on the final design, a minimized audit record may contain an application reference, a rule-set version, an assessment result, a decision reference, a timestamp, and a cryptographic hash. However, the actual audit payload must be documented from the implemented system rather than inferred from conceptual architecture. Cryptographic hashing can help detect whether data has changed when a recomputed digest is compared with a previously recorded digest [7]. Nevertheless, a hash does not encrypt the original data, prove the

truthfulness of an application, or independently establish that identity verification was performed correctly. The design rationale is therefore to use storage and audit mechanisms for their respective purposes: protecting sensitive application information and preserving appropriate evidence of processing events.

Table 2 : Comparison of existing approaches and the proposed research

Approach	**Main capability**	**Limitation relevant to this research**
Decentralized identity standards	Define interoperable identity-related concepts	Do not provide the complete welfare workflow
Facial and document verification	Provide evidence for identity checking	Require reliable verification procedures and evaluation
Conventional welfare assessment	Applies programmed-specific eligibility criteria	Consistency depends on the procedures and implementation
Smart-contract-based assessment	Execute predefined business rules	Cannot guarantee that rules or input data are correct
Blockchain audit trail	Records transactions for later examination	Does not independently guarantee privacy, fairness, or truthful inputs
Proposed SmartGrama component	Connects identity verification, officer review, welfare-rule assessment, and final decision-making	Requires evidence that the implemented workflow behaves as intended

The comparison demonstrates that each existing approach addresses only part of the overall problem. Decentralized identity standards establish identity-related concepts, facial and document verification provide evidence for identity checking, welfare

assessment techniques apply programmed-specific eligibility criteria, smart contracts execute predefined business rules, and blockchain audit trails support the examination of recorded transactions. The SmartGrama research focuses on integrating these capabilities into a coherent workflow while maintaining clear responsibilities, preserving the authority of the responsible officer over the final welfare decision, and ensuring appropriate handling of sensitive information. The effectiveness of the integrated component must ultimately be established through evidence that the implemented workflow behaves as intended.

2.5 Gap Synthesis and Conceptual Framework

The reviewed literature establishes the technical foundations of decentralized identity, biometric verification, welfare targeting, smart-contract execution, and blockchain auditability. However, the availability of these individual capabilities does not, by itself, demonstrate how they can be integrated into a welfare-service workflow that maintains human oversight and protects sensitive information. The research gap addressed by this component is therefore the design, implementation, and evaluation of an integrated identity-verification and welfare eligibility-assessment workflow.

One major research gap concerns the integration of identity verification and welfare assessment. Identity verification determines whether the applicant's evidence supports the identity being claimed, whereas eligibility assessment determines whether the application satisfies the requirements of a particular welfare programmed. Although these processes are related, they address different questions. The SmartGrama component connects identity verification with the welfare application process to establish a structured workflow from existing identity information to eligibility assessment and final decision-making.

Another research gap concerns the appropriate use of automated checks and human oversight. Automated facial and document checks can support identity verification, while software-based rules can apply eligibility criteria consistently. However, automated results may require further examination when evidence is inconsistent, or an application presents an exceptional case. The proposed workflow incorporates

authorized officer review into the identity-verification process and preserves the officer's authority over the final welfare decision.

Privacy-conscious auditability represents a further research gap because welfare applications may contain sensitive personal, financial, household, and biometric information. An auditable workflow should preserve appropriate evidence of processing without exposing unnecessary personal information through the audit mechanism. Accordingly, the research considers separating application evidence from audit information and using suitable references and integrity checks to support traceability while minimizing unnecessary exposure of sensitive data.

The need for systematic evaluation is also an important research gap. Technically plausible architecture does not automatically establish accuracy, fairness, efficiency, or improved citizen trust. These outcomes require evidence from appropriate testing and evaluation procedures. Therefore, the research evaluates the component against its implemented functionality, including identity-verification scenarios, welfare eligibility-rule outcomes, officer decisions, and relevant audit or integrity checks were supported by the prototype and available test evidence. Collectively, these gaps lead to the central research problem: how can identity verification, welfare eligibility-rule execution, authorized human decision-making, and privacy-conscious auditability be integrated into a coherent and testable welfare-service workflow?

The proposed conceptual framework illustrates the relationship between identity verification, officer review, welfare application processing, smart-contract eligibility assessment, and final decision-making. As presented in Figure 2.1, the workflow begins with the existing SmartGrama identity and its registered identity information and reference. The identity-verification stage uses facial verification and NIC or other document evidence, after which an authorized officer reviews the verification evidence and records the relevant decision. The applicant then proceeds through the welfare application process, where the information required for the selected welfare programmed is provided. The smart-contract eligibility-assessment stage executes predefined eligibility rules and produces an assessment result, which supports the responsible officer's final welfare decision. Privacy and data minimization, authorized

access, traceability, and integrity checking are cross-cutting requirements throughout the workflow.



Figure 1 : Conceptual Framework for Privacy-Preserving Decentralized Identity Verification and Smart-Contract-Based Welfare Eligibility Assessment.

The conceptual framework consists of three principal stages. The first establishes identity-related evidence through the verification process and authorized review. The second assesses the welfare application against predefined eligibility rules. The third preserves the responsible officer's authority to make the final decision. Privacy and auditability apply across all stages rather than being limited to a single process. The framework represents the intended conceptual workflow, and the final thesis

illustration should reflect the actual implementation. In particular, it should not imply that a smart contract itself performs facial recognition or document verification unless these functions are demonstrably part of the implemented contract.

The relationship between the identified research gaps and the evaluation process is summarized in Table 3. This alignment connects the research questions with observable system behavior and identifies the evidence required to assess whether the component addresses the problems identified in the literature.

Table 3 : Research gap and evaluation alignment

Research gap	**Research question**	**Evaluation focus**
Integration of identity verification and welfare assessment	How can identity verification and eligibility assessment be connected in one workflow?	Correct workflow progression and handling of verification outcomes
Automated checks and human oversight	How should automated verification and eligibility results be incorporated into officer decisions?	Valid and invalid evidence, assessment outcomes, and officer approval or rejection
Privacy-conscious auditability	How can processing events be traced while minimizing unnecessary exposure of sensitive data?	Data separation, audit-record contents, and integrity verification
Need for implementation evidence	Does the implemented prototype behave as intended under controlled scenarios?	Test coverage, rule consistency, error handling, and measured results where available

This alignment ensures that the research questions are connected to observable behavior rather than broad assumptions about the benefits of blockchain or biometric technology. It also establishes a basis for evaluating the implemented component

through defined scenarios, appropriate evidence, and measurable results where available.

This chapter reviewed the foundations and related research relevant to privacy-preserving decentralized identity verification and smart-contract-based welfare eligibility assessment. It examined decentralized identity standards, biometric and document-based verification, Sri Lankan welfare-targeting research, smart-contract execution, and privacy-conscious audit design. The review identified the need to integrate these capabilities into a coherent workflow while distinguishing identity verification from eligibility assessment and preserving authorized officer decision-making. It also established that the accuracy and effectiveness of the component must be demonstrated through appropriate testing rather than assumed from the technologies used. The proposed conceptual framework connects the existing SmartGrama identity, verification evidence, officer review, welfare application, smart-contract eligibility assessment, and final decision. The next chapter describes the research methodology used to develop and evaluate the component, including the research design, test scenarios, evaluation measures, data analysis, privacy considerations, and reproducibility.

3 RESEARCH METHODOLOGY

3.1 Research design

This research adopted a design-and-development approach to investigate the integration of identity verification and welfare eligibility assessment within the SmartGrama digital platform. The research involved defining the component requirements, designing the workflow, developing the prototype, and evaluating its behavior through controlled test scenarios. The approach focused on connecting several related functions, including the use of an existing digital identity, identity-verification evidence, authorized officer review, welfare eligibility-rule assessment, and the recording and verification of relevant processing events. The resulting component was designed to support a structured welfare-service workflow while preserving the officer's responsibility for the final decision.

The development process involved identifying the functional and non-functional requirements of the individual component, designing the workflow connecting identity verification and welfare eligibility assessment, implementing the relevant component interfaces and processing logic, developing the welfare eligibility-assessment and audit-related functionality within the scope of the prototype, testing implemented functions using controlled inputs and defined expected outcomes, and analyzing the resulting test evidence to identify successful behavior, limitations, and areas requiring further evaluation. The research combined software engineering activities with experimental testing. The software engineering activities established the design and implementation of the component, while the testing activities examined whether the implemented functions behaved according to their defined requirements.

The research used a prototype-based strategy because its objective concerned the design and evaluation of a technical workflow rather than a survey of citizen opinions or a statistical study of a population. The prototype provided a means of examining how identity-related verification, eligibility-rule execution, officer decisions, and audit-related functions could be connected. Controlled inputs were used to exercise the relevant processing paths without requiring the collection of real citizens' welfare or biometric information. The strategy was informed by the technical concepts discussed in Chapter 2, including decentralized identity, biometric verification, rule-based

assessment, smart contracts, and cryptographic integrity checking. The selected technologies were considered in relation to the component's requirements rather than on the assumption that any single technology automatically guarantees accuracy, security, or fairness.

The relationship between the research objectives, development activities, and evaluation methods is summarized in Table 4.

Table 4 : Research design and evaluation alignment

Research focus	**Development activity**	**Evaluation approach**
Identity verification	Design and implement the identity-verification workflow within the component scope	Controlled verification scenarios and expected outcome checks
Welfare eligibility assessment	Implement the defined assessment rules and associated processing logic	Rule-based test cases, including valid, invalid, and boundary inputs
Officer oversight	Represent the review and final decision stages in the workflow	Check that the relevant decision states and transitions behave as specified
Auditability and integrity	Implement the audit-related functions supported by the prototype	Inspect recorded audit data and verify hashes or record relationships were implemented
Privacy and security	Define data-handling boundaries and minimize unnecessary exposure	Review data fields, access boundaries, and the handling of sensitive information
Component integration	Define and test interfaces between relevant services	Interface and integration tests within the available test environment

The evaluation focused on observable behavior and implementation evidence. Each test was intended to compare the actual output or state change with the expected result defined by the corresponding requirement. This approach provided a structured basis for examining whether the implemented component functions operated as intended and for identifying limitations that required further investigation. The scope of the evaluation was limited to the individual research component. It did not attempt to validate the entire SmartGrama platform or independently establish the correctness of the national welfare policy.

3.2 Data, Participants, or Study Materials

3.2.1 Study Materials

The study used software artefacts, technical specifications, component requirements, workflow designs, existing SmartGrama interfaces, eligibility rules, audit functions, and available test reports. These materials supported the development and evaluation of the identity-verification and welfare eligibility-assessment component without requiring the collection of real citizens' personal information.

3.2.2 Test Data

Controlled test data were used to evaluate identity verification, welfare applications, eligibility assessment, officer decisions, and audit operations. The test scenarios included valid and invalid identity evidence, inconsistent document information, eligible and ineligible applications, boundary conditions, and officer approval or rejection. The data were intended to test software behavior under defined conditions rather than measure real-world welfare eligibility or biometric accuracy. The actual data sources and composition should be documented using implementation records.

Table 5 : Categories of test data used in the research

Data category	**Purpose**	**Important consideration**
Identity references	Representing an applicant's existing identity	Use controlled identifiers

Facial-verification inputs	Test supported verification scenarios	Workflow tests do not establish biometric accuracy
NIC/document information	Test valid, invalid, or inconsistent documents	Distinguish validation from simple field checking
Welfare application data	Provide eligibility-assessment inputs	Use implemented criteria
Eligibility-rule parameters	Define assessment conditions	Record rule versions and boundaries where applicable
Officer decisions	Represent review, approval, or rejection	Separate final decisions from automated results
Audit and integrity data	Test record retrieval and hash verification	Use implemented mechanisms only

3.2.3 Participants and Human Subjects

The evaluation focused on software behavior and controlled test cases. No study involving real welfare applicants, citizen surveys, or actual officer interviews is claimed. The authorized officer was represented as a system role. Therefore, the findings can demonstrate tested software behavior but cannot independently establish user satisfaction, institutional acceptance, or improved access to welfare services.

3.3 Experimental or Study Procedure

3.3.1 Overview of the Procedure

The research followed six main activities: requirements analysis, workflow and architecture design, prototype implementation, test-case preparation, test execution, and evidence analysis. This sequence established the component's scope, developed the intended workflow, and provided a structured method for comparing actual system behavior with expected outcomes.



Figure 2 : Research and evaluation procedure for the individual SmartGrama component.

Figure 3.1 illustrates the methodological sequence. The final figure should represent the activities actually completed during development and testing.

3.3.2 Requirements Analysis

The component requirements were identified according to their responsibilities within SmartGrama, focusing on identity verification and welfare eligibility assessment. Functional requirements covered verification processing, application assessment, officer decisions, and audit operations where implemented. Non-functional requirements addressed privacy, security, consistency, maintainability, traceability, and separation of responsibilities. These requirements formed the basis for the test cases.

3.3.3 Component Design and Implementation

The component connected identity-related evidence, predefined eligibility rules, and officer decision-making. The implementation was evaluated according to the functions available in the prototype, distinguishing application workflows from actual biometric algorithms and external service integrations. For example, testing a verification endpoint does not prove that facial matching occurred, and testing a mock blockchain provider does not demonstrate successful transactions on a live Hyperledger Fabric network.

3.3.4 Identity-Verification Test Procedure

Identity-verification testing examined valid matching evidence, facial mismatches where actual comparison was supported, invalid or inconsistent NIC/document information, officer review, and approval or rejection. Each test should record its inputs, expected result, actual result, and status. Simulated verification responses must be reported as workflow tests rather than biometric-accuracy measurements. Invalid or inconsistent evidence should not automatically be treated as successful verification.

3.3.5 Welfare Eligibility-Assessment Test Procedure

Eligibility testing examined applications that satisfied or failed the configured rules, boundary values, different combinations of implemented eligibility factors, and officer decisions. Actual outputs were compared with expected results derived from the implemented rules. The tests evaluated rule consistency, not whether the rules represented an officially validated national welfare-assessment methodology.

3.3.6 Audit and Integrity Verification Procedure

Audit testing examined record creation, retrieval, and integrity verification where implemented. For SHA-256, unchanged data processed consistently should produce the same digest, while modifications to the hashed content should produce a different digest with overwhelming probability. Where a mock blockchain provider was used, the findings were limited to the audit service and mock-provider behavior rather than live blockchain execution.

3.3.7 Test Execution and Evidence Recording

Tests were conducted using the available prototype environment. Evidence included test identifiers, inputs, expected and actual outputs, pass/fail status, errors, logs, and screenshots where available. Only executed tests should be reported as completed and failed or inconclusive results should be retained for analysis. This evidence supports the results presented in Chapter 6.

3.4 Measures and Evaluation Metrics

3.4.1 Purpose of Evaluation

The evaluation measures focused on functional correctness, identity-verification behavior, eligibility-rule consistency, officer-decision handling, and audit integrity where supported. Each measure was selected to assess an observable property of the implemented component without assuming improvements in real-world welfare outcomes.

3.4.2 Functional Test Pass Rate

The functional test pass rate measures the percentage of executed tests that produced the expected results.

A test passes when its actual result matches its expected result. Planned but unexecuted tests must not be counted as successful. This metric summarizes test outcomes but does not prove complete test coverage or the absence of defects.

3.4.3 Identity-Verification Test Outcomes

Identity-verification tests examined the scenarios supported by the prototype. Where actual biometric predictions and independently established labels were available, accuracy could be calculated as follows:

\[\text{Accuracy}= \frac{TP+TN}{TP+TN+FP+FN} \]

Here, \(TP\), \(TN\), \(FP\), and \(FN\) represent true positives, true negatives, false positives, and false negatives. This metric must only be used when genuine classification results and reliable ground-truth labels are available. Otherwise,

evaluation should focus on verification states, error handling, and officer-review transitions.

3.4.4 Eligibility-Rule Consistency

Eligibility-rule consistency measures whether the implemented rules produce the expected results for defined inputs.

Expected results must be derived from the configured rules. This metric measures implementation correctness rather than the suitability of the rules for national welfare policy. Boundary testing is particularly important for criteria involving thresholds.

3.4.5 Audit Integrity and Traceability

Audit evaluation examined whether records were created and retrieved correctly, whether unchanged data passed hash verification, whether modifications were detected, and whether supported record relationships were handled correctly. These tests assess the implemented audit mechanism but do not establish the availability, consensus behavior, or transaction finality of a live blockchain network when a mock provider is used.

3.4.6 Evaluation Metrics Summary

Table 6 : Evaluation of metrics and evidence requirements

Metric	**Purpose**	**Required evidence**
Functional test pass rate	Measure test success	Executed tests and pass/fail results
Verification scenario outcomes	Assess workflow behavior	Inputs, expected and actual states, and errors
Biometric accuracy, where applicable	Measure classification performance	Actual matching outputs and labelled test data
Eligibility-rule agreement	Compare rules with expected outcomes	Configured rules and expected results

Audit integrity verification	Check detection of data changes	Original records, hashes, modified records, and outputs
Record retrieval and traceability	Assess record access and relationships	API outputs, identifiers, and logs
Processing time, where measured	Measure operation duration	Reliable timestamps or instrumentation

These metrics provide measurable evidence of implemented behavior while avoiding unsupported conclusions about overall system effectiveness.

3.5 Data Analysis

3.5.1 Analysis of Functional Test Results

Actual outputs were compared with expected outcomes, and each executed test was classified as passed, failed, or inconclusive. Failed tests were examined for possible causes, such as incorrect rule conditions, invalid state transitions, unexpected responses, data-handling errors, or connected-service failures. A specific cause should only be reported when supported by evidence. Chapter 6 should present the executed, passed, and failed test counts and identify skipped or inconclusive cases separately.

3.5.2 Analysis of Eligibility Assessment

Eligibility outputs were compared with expected results derived from the configured rules, including qualifying and non-qualifying applications and boundary conditions. The analysis distinguished automated assessment results from final officer decisions because a correct assessment does not automatically mean that an application has been approved. Findings were limited to the conditions covered by the executed tests.

3.5.3 Analysis of Identity-Verification Behavior

Identity-verification results were examined according to the functionality implemented. Workflow testing focused on verification states, document information, review transitions, and approval or rejection outcomes. Accuracy could only be assessed where genuine biometric or document-validation outputs and appropriate

expected labels were available. A successful API response alone was not considered proof of verification accuracy.

3.5.4 Analysis of Audit and Integrity Behavior

Audit outputs were examined to determine whether records were created, retrieved, and verified as expected. Where tested, hash verification results for unchanged and modified data were compared. Successful hash verification was not interpreted as proof that the original information was truthful or that all forms of tampering could be detected. Results from a mock provider were limited to the behavior of the tested audit service.

3.5.5 Interpretation and Reporting

The research used descriptive summaries and test evidence rather than population-level statistical inference. Quantitative metrics summarized executed tests and measured timings, while logs and failure analysis helped explain implementation limitations.

3.6 Ethics, Privacy, Security, and Responsible AI

3.6.1 Data Minimization and Privacy Protection

The research considered data minimization by limiting information collection, access, and retention to what was required for the relevant processing purpose. Controlled test data should be preferred over real citizen records where possible, and personal or biometric evidence should be separated from audit information with appropriate access controls. The actual audit payload must be checked to establish whether unnecessary personal information has been excluded.

3.6.2 Security and Integrity

Security considerations included identity-data handling, input validation, access controls, and integrity checking where implemented. Cryptographic hashing can help detect changes to data but does not provide confidentiality or prove that the original data were valid. Confidentiality requires appropriate access controls and, where necessary, encryption. The thesis should distinguish the component's own controls from those provided by other SmartGrama services and infrastructure.

3.6.3 Responsible Use of Biometric Verification

Facial verification may be affected by false acceptance, false rejection, image quality, and demographic differences in algorithm performance. NIST research highlights the importance of considering these limitations [3]. SmartGrama therefore treats verification as part of a broader process involving authorized officer review. However, effective oversight still depends on clear procedures, access to relevant evidence, and appropriate handling of uncertain or conflicting results.

3.6.4 Fairness and Welfare Decision-Making

Welfare eligibility decisions can significantly affect individuals and households. Eligibility rules should therefore be explicit, consistently implemented, and aligned with applicable programmed requirements. The research distinguishes between executing rules correctly and establishing whether those rules are fair, legally appropriate, or socially beneficial. The officer responsible retains the final decision-making role, and the component should not be presented as independently determining legal entitlement to welfare benefits.

3.6.5 Limitations of the Ethical Evaluation

This section establishes ethical and security considerations but does not claim that a formal privacy-impact assessment, independent security audit, demographic fairness study, or institutional ethics review was completed unless supporting evidence exists. Any participant involvement, consent procedures, institutional approvals, or use of real personal information must be accurately documented. If only controlled software test data were used, this limitation should be stated clearly.

3.7 Reproducibility and Research Artefacts

3.7.1 Reproducibility of the Evaluation

Reproducibility requires sufficient information about the implementation and test environment for another researcher to understand and, where possible, repeat the evaluation. The thesis should document the development environment, software versions, configuration, test commands, inputs, expected outputs, and execution procedure using actual project records. Test cases should define the conditions for

success and failure, particularly because changes to eligibility rules can change expected outcomes.

3.7.2 Research Artefacts

The principal research artefacts include component source code, requirements, workflow and architecture diagrams, test-case definitions, test results, and relevant API documentation. Additional artefacts may include audit records, hash-verification outputs, configuration files, and screenshots. Sensitive credentials, private keys, and real citizen data should not be included in submitted artefacts.

3.7.3 Versioning and Documentation

The thesis should identify the exact component version evaluated and distinguish it from earlier prototypes or later team integration. It should also identify functions tested through local or mock services and those tested through actual integrated services. This distinction helps ensure that the findings are interpreted accurately and provides a basis for future evaluation.

3.7.4 Chapter Summary

This chapter described the prototype-based design-and-development methodology used to evaluate the SmartGrama identity-verification and welfare eligibility-assessment component. It summarized the study materials, test data, research procedure, evaluation metrics, data-analysis methods, ethical considerations, and reproducibility requirements. The methodology emphasizes controlled testing and distinguishes implemented functions from capabilities requiring further evidence..

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

4.1.1 Introduction

The SmartGrama privacy-preserving decentralized identity verification and welfare eligibility assessment component was designed to provide a structured, explainable, and auditable process for evaluating welfare applications. It combines identity verification, Know Your Customer (KYC) review, configurable welfare eligibility assessment, authorized officer decision-making, and blockchain-based audit recording.

The requirements were derived from the research problem, component objectives, and proposed SmartGrama workflow. The design prioritizes privacy, consistency, traceability, and responsible automated decision support. As welfare applications contain sensitive personal and household information, detailed assessment data are separated from the minimum information required for auditability.

The eligibility assessment considers six dimensions: education, health and special needs, economic level and income, assets and ownership, housing conditions, and family demography. The assessment engine generates a prototype score, category, and explanation using configurable rules. These outputs support the authorized officer's review rather than automatically determining final welfare entitlement.

4.1.2 Functional Requirements

Functional requirements define the main operations supported by the component.

Table 7 : Functional Requirements

ID	Requirement	Description
FR01	Identity association	Associate a welfare application with an existing SmartGrama digital identity reference.
FR02	Identity verification	Support facial verification and NIC or other document-based verification workflows.

FR03	KYC review	Enable an authorized officer to review submitted verification information.
FR04	Welfare application	Capture household information required for the selected assessment.
FR05	Eligibility calculation	Apply configurable rules across the six welfare assessment dimensions.
FR06	Explainable assessment	Generate a prototype score, category, and explanation.
FR07	Human decision-making	Require an authorized officer to make the final approval or rejection decision.
FR08	Rule versioning	Associate each assessment with the applicable rule configuration or version.
FR09	Audit record creation	Construct a privacy-minimized audit payload for traceability.
FR10	Integrity verification	Use SHA-256 hashing to help detect changes to canonicalized audit data.
FR11	Reassessment history	Support traceability of subsequent assessments and decisions.
FR12	Blockchain interface	Provide an interface for submitting and verifying audit records through the intended permissioned blockchain integration.

These requirements describe the responsibilities of the individual component. The behavior of shared identity services and the group's blockchain infrastructure depends on the relevant integration interfaces.

4.1.3 Non-functional Requirements

Non-functional requirements define the expected quality and operational constraints of the component.

Table 8 : Non Functional Requirements

ID	Requirement	Design consideration
NFR01	Privacy	Avoid storing complete identity documents, biometric data, or household records directly on-chain without an authorized justification.
NFR02	Security	Restrict sensitive operations and officer decisions to authorized users and services.
NFR03	Integrity	Use cryptographic hashing to detect changes to audit data.
NFR04	Explainability	Provide meaningful assessment explanations rather than only a score.
NFR05	Reliability	Validate inputs and handle invalid, incomplete, or inconsistent data safely.
NFR06	Maintainability	Separate eligibility rules, assessment logic, decision handling, and audit functions.
NFR07	Traceability	Preserve references to applications, rule versions, decisions, and audit events.
NFR08	Interoperability	Define clear interfaces for identity, frontend, and blockchain integration.
NFR09	Accountability	Preserve the authorized officer's final decision-making responsibility.
NFR10	Testability	Support controlled testing of normal cases, invalid inputs, reassessment, and audit integrity.

The requirements priorities correctness and responsible handling of sensitive information. Although a permissioned blockchain can support controlled participation and traceability, it cannot independently guarantee the accuracy of application data or the fairness of eligibility rules.

4.1.4 Constraints and Assumptions

The component was developed within the scope of an undergraduate research prototype. Its eligibility parameters are configurable research settings and should not be interpreted as the official Aswesuma eligibility formula. Operational use would require validated policy rules, authorized access, appropriate governance, and approval by the responsible authorities.

The reported software tests used controlled or synthetic records rather than real citizens' welfare records. The design assumes that the wider SmartGrama platform can provide identity references and that authorized services can support the required integrations. The precise deployment status and responsibilities of the blockchain infrastructure must be described according to the available implementation evidence.

4.2 Component Architecture within the Complete Solution

4.2.1 Architectural Overview

The component follows a modular architecture that separates identity context, welfare assessment, human decision-making, and audit integrity. This separation enables individual functions to be developed and tested independently while maintaining a defined workflow.

Within the wider SmartGrama platform, the component focuses on identity verification, welfare eligibility assessment, officer decisions, and audit information. It does not claim ownership of the entire registration system or every blockchain data storage function.

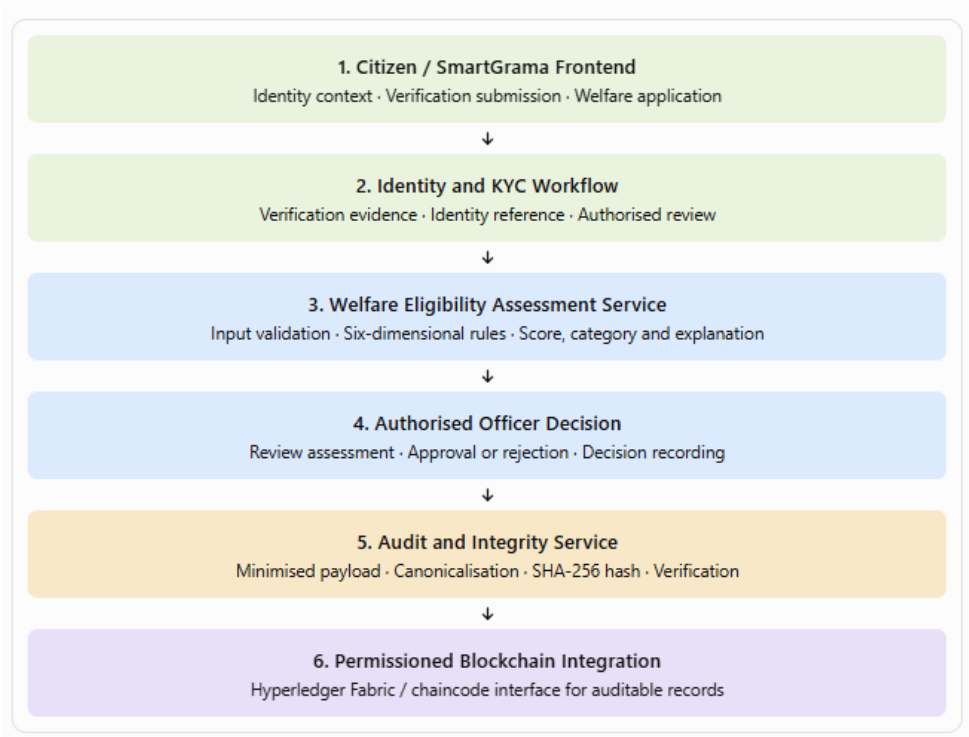


Figure 3 : High-level architecture of the SmartGrama identity and welfare eligibility component

4.2.2 Data Flow and Module Interaction

The component follows a sequence of identity verification, welfare assessment, officer review, and audit recording.

1. The citizen uses an existing SmartGrama identity.
2. Identity evidence is submitted through the verification workflow.
3. An authorized officer reviews the verification information.
4. The verified identity reference is associated with a welfare application.
5. The system validates the six-dimensional welfare inputs and applies the configured rules.
6. The assessment result and explanation are presented to the officer.
7. The officer records an approval or rejection decision.

8. The system constructs a minimized audit payload and calculates its integrity hash.
9. Audit information is submitted through the blockchain integration interface.
10. The system supports subsequent audit verification and reassessment traceability.

4.3 Design Alternatives and Decisions

The design decisions considered the sensitivity of welfare data, assessment explainability, human accountability, and the scope of the research prototype.

Design concern	**Alternative**	**Selected design and justification**
Identity management	Re-enter identity details for every application	Reference an existing SmartGrama identity to reduce duplication.
Eligibility assessment	Manual assessment only	Use configurable rules for consistent calculations while retaining officer review.
Decision authority	Fully automated approval or rejection	Retain the authorized officer's final decision.
Rule configuration	Hard-coded parameters throughout the application	Separate configurable parameters and rule versions for maintainability and traceability.
Audit storage	Store complete applications on chain	Use a minimized audit payload to reduce exposure of personal information.
Integrity protection	Rely only on application database records	Use SHA-256 hashing alongside appropriate access controls and record management.
Blockchain model	Public blockchain	Use a permissioned approach to support-controlled participation.

Blockchain dependency	Couple all modules directly to ledger functions	Use a provider interface to isolate ledger integration from assessment logic.

4.3.1 Reuse of Existing Digital Identity

Creating a new identity for every welfare transaction would duplicate identity information and complicate application-history management. The selected design uses an existing SmartGrama identity reference to connect verification with the welfare workflow. This reduces repeated identity-data handling but depends on the reliability and availability of the shared identity service.

4.3.2 Configurable Eligibility Assessment

Manual assessment offers flexibility but can produce inconsistent calculations. Fully automated decisions can apply rules consistently but may not adequately address exceptional circumstances. The selected approach generates an assessment result and explanation using configurable rules while retaining the officer's final decision-making authority.

4.3.3 Privacy-aware Audit Storage

Storing complete identity and welfare records on a blockchain could expose sensitive information persistently across ledger participants. The design therefore separates detailed application data from the minimum audit information needed to trace decisions. Where appropriate, references, decision metadata, and hashes can be recorded while detailed personal information remains in controlled off-chain storage.

A hash is not encryption. Hashing predictable or low-entropy personal information may still permit inference. Data minimization, access controls, and careful selection of audit fields are therefore necessary.

4.3.4 Permissioned Blockchain and Provider Abstraction

A public blockchain may conflict with the privacy and governance requirements of welfare services. A permissioned model such as Hyperledger Fabric is more consistent

with controlled organizational participation. The provider interface also reduces dependence on a specific ledger implementation.

4.4 Detailed Design

4.4.1 Welfare Assessment Data Model

The assessment model uses six dimensions identified as the SmartGrama component.

Table 9

Dimension	**Purpose in the prototype**
Education	Represents education-related information used in the configured assessment.
Health and special needs	Represents relevant health-related or special-needs information.
Economic level and income	Represents household economic circumstances and income indicators.
Assets and ownership	Represents selected household asset and ownership indicators.
Housing conditions	Represents relevant housing characteristics.
Family demography	Represents household composition and demographic characteristics.

Each dimension contributes according to the configured prototype rules. The assessment service validates the input, applies the active configuration, and generates a result. The exact variables, weights, thresholds, and category boundaries must be documented from the implemented configuration rather than invented.

4.4.2 Eligibility Calculation and Explanation

The eligibility engine processes validated inputs through the following stages:

1. Validate required fields and their formats.
2. Load the applicable rule configuration and version.

3. Evaluate the configured rules across the six dimensions.
4. Calculate the prototype score or configured result.
5. Map the result to the relevant prototype category.
6. Generate an explanation of the assessment factors.
7. Return the result to the officer-review workflow.

The score and category are decision-support outputs, not substitutes for official eligibility policy or the officer's final decision. Retaining the rule-version reference also helps reviewers identify the configuration used for each assessment.

4.4.3 Officer Decision Workflow

After the assessment is generated, the authorized officer reviews the application information and system output before recording an explicit decision, such as APPROVED or REJECTED.

The audit event should distinguish the system-generated assessment from the officer's final decision. This distinction supports accountability because an assessment category is not necessarily equivalent to official approval. The event should also identify the relevant application and rule version, subject to privacy and access requirements.

4.4.4 Audit Payload and Integrity Verification

The audit module creates a reduced representation of the decision event.

Field	**Purpose**
Application reference	Identifies the welfare application without embedding the full application.
DID reference	Connects the audit event to the relevant digital identity reference.
Rule version	Identifies the configuration used for assessment.
Assessment result	Records the relevant prototype outcome.
Officer decision	Records the authorized officer's final decision where required.

Timestamp	Records of when the event occurred.
Integrity hash	Supports verification that canonicalized audit data have not changed.

The exact field names and on-chain/off-chain placement must match the implemented audit schema. This table describes the design and does not claim that every field is stored on-chain.

The integrity process prepares a consistent representation of the audit data, canonicalizes it, calculates a SHA-256 digest, and associates the digest with the audit event. During verification, the digest is recalculated and compared with the recorded value. A mismatch indicates that the data or their representation may have changed.

4.4.5 Reassessment and Audit History

Welfare circumstances may change, requiring an application to be reassessed. The design retains references to the application, rule configuration, and decision events so that subsequent assessments can be distinguished from earlier ones.

A reassessment should not silently overwrite historical decisions. Instead, the system should preserve relationships between earlier and later events according to the capabilities of the implemented audit model. This supports traceability and helps reviewers identify the assessment and decision relevant to a particular stage.

4.4.6 Error Handling and Validation

The component should handle invalid inputs, missing fields, unsupported rule configurations, duplicate audit submissions, and integrity-verification failures explicitly. Validation messages should avoid exposing unnecessary personal information.

Blockchain-operation failures must also be distinguished from assessment failures so that the system can report the correct status and support appropriate recovery. These behaviors should be described as implemented only when supported by source code and test results; otherwise, they should be identified as design requirements or future improvements.

4.5 Standards and Professional Practice

4.5.1 Privacy and Data Minimization

The component process’s identity references and household information, so it follows the principle of collecting, processing, and sharing only the information necessary for each operation. Detailed personal and biometric information should not be included in blockchain records without a clear, authorized need. Access to identity evidence, household records, and officer decisions should be controlled according to users' responsibilities.

The final implementation should be assessed against applicable Sri Lankan data-protection legislation and institutional policies. Data minimization alone does not establish legal compliance.

4.5.2 Cryptographic Integrity

SHA-256 is used as a cryptographic hash function for audit-data integrity checking. Comparing recalculated and recorded digests can help identify modifications. However, hashing is not encryption or a complete security mechanism. Where applicable, integrity checking should be supported by access controls, credential management, reliable timestamps, and controlled audit access.

4.5.3 Blockchain Governance and Access Control

A permissioned blockchain requires defined participants, access permissions, and governance arrangements. The design must consider which organizations or services can submit audit events, who can read relevant records, and how exceptional situations are handled. The actual governance model should be documented according to the group's blockchain implementation rather than assumed solely from the use of Hyperledger Fabric.

4.5.4 Responsible Automated Assessment

The eligibility engine supports administrative assessment without removing human accountability. It should provide understandable results, apply configured rules consistently, and identify the configuration used for each assessment.

The six dimensions and their parameters require appropriate review for relevance and fairness before operational use. Prototype test success does not establish that the rules are socially fair, legally approved, or suitable for real-world welfare decisions.

4.5.5 Professional Software Engineering Practice

The modular design supports separation of concerns, maintainability, and testability. Requirements are linked to assessment, officer decision, and audit operations, allowing controlled tests of normal behavior, invalid inputs, reassessment, and audit integrity.

Version control, code review, reproducible configuration, and documented interfaces should be followed where applicable. The thesis must distinguish practices followed during development from recommendations for future production deployment.

4.5.6 Standards and Guidance Relevant to the Design

The following sources are relevant to the design and should be included in the thesis reference list after verifying their exact editions and citation details:

- W3C Verifiable Credentials Data Model: Relevant to representing verifiable identity claims and limiting repeated disclosure of identity information.
- Hyperledger Fabric documentation: Relevant to permissioned blockchain architecture, identities, access control, and smart-contract or chain code concepts.
- NIST Secure Hash Standard (FIPS PUB 180-4): Relevant to SHA-256 and its cryptographic properties.
- Applicable Sri Lankan data-protection legislation and institutional policies: Relevant to the processing and protection of personal information.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development Environment and Technology Stack

The SmartGrama welfare eligibility and blockchain audit component was developed as a modular component of the wider SmartGrama platform. Its backend services support welfare eligibility assessment, officer decisions, audit-record generation, and blockchain interaction. A provider abstraction separates application logic from blockchain-specific operations, enabling independent development and testing.

The component uses SHA-256 hashing to support audit-record integrity verification. Before submission, audit data are prepared in a consistent format and hashed. The digest can later be recalculated to identify changes to the data.

Table 10 : Technology Stack and Implementation Purpose

Technology or concept	**Role in the component**
TypeScript and Node.js	Backend service implementation and execution
Modular service architecture	Separates eligibility, decision, and audit responsibilities
Provider interface	Separates application services from blockchain implementations
Hyperledger Fabric	Permissioned blockchain environment
Smart contract or chain code	Manages blockchain-side audit operations
SHA-256	Generates cryptographic digests for integrity verification
Automated tests	Verifies service logic, eligibility behavior, and audit operations
REST API	Supports communication between the backend and connected components

Development involved application-level testing and blockchain integration. Mock-provider testing supported audit-logic development, while the completed live Fabric

integration enabled communication with the configured blockchain environment. Exact software versions and configurations should be verified before submission.

5.2 Component Implementation

The component was implemented around five main responsibilities: eligibility assessment, officer decision-making, audit-payload preparation, blockchain record management, and integrity verification.

5.2.1 Welfare Eligibility Assessment

The eligibility service processes application information across six dimensions: education, health and special needs, economic level and income, assets and ownership, housing conditions, and family demography.

Configurable assessment parameters are applied to generate a prototype score and category. Separating these parameters from the assessment logic supports maintainability and allows the rules to be revised independently. The calculation is a research-prototype mechanism, not a reproduction of the official production Aswesuma Proxy Means Test algorithm.

5.2.2 Officer Decision Workflow

The officer workflow separates the calculated assessment from the final administrative decision. The result is presented for review, after which the authorized officer records the decision.

This distinction ensures that a prototype score or category is not treated as official welfare approval or rejection. The recorded decision and application references can subsequently support audit-record preparation.

5.2.3 Audit-Payload Preparation

The audit service prepares a minimized payload for blockchain recording, avoiding unnecessary duplication of the complete citizen dossier. Depending on the implemented schema, it may contain an audit-record identifier, application identifier, DID reference, welfare scheme, rule version, assessment category, eligibility status, record status, timestamp, previous-record reference, and payload hash.

Identity-document images, biometric images, bank-account details, and unnecessary household information should remain outside the blockchain payload. The actual transaction data must be checked before making claims about privacy compliance.

5.2.4 Cryptographic Integrity Verification

The audit service generates a SHA-256 digest from the relevant payload after normalizing its representation. Consistent formatting helps ensure that differences in property ordering or serialization do not produce unnecessary hash changes.

During verification, the digest is recalculated and compared with the expected hash. A mismatch indicates that the data does not match the expected representation. Hashing supports integrity checking but does not encrypt information or prove the accuracy of the original data.

5.2.5 Blockchain Record Management

The blockchain provider abstraction supports audit-record creation, retrieval, and integrity verification without embedding network-specific communication logic throughout the application.

Initially, the component used a mock blockchain provider for controlled testing. The live Hyperledger Fabric integration has subsequently been completed, enabling communication with the configured Fabric environment. The mock provider remains useful for isolated testing, but its simulated operations must be distinguished from transactions executed through the live provider.

5.3 Integration with the Other Components

The component forms part of the wider SmartGrama platform. Its integration responsibilities include associating application information, processing eligibility results, recording authorized decisions, and linking audit evidence to the relevant application.

5.3.1 Backend Service Integration

The eligibility and audit services communicate through defined interfaces. This allows assessment results to be reviewed and used for audit preparation while supporting

independent testing. Eligibility calculations can be tested without submitting every test to the blockchain, and audit operations can be examined using controlled payloads.

5.3.2 Live Hyperledger Fabric Integration

The completed Fabric integration connects the backend blockchain provider to the configured permissioned network. The application prepares the audit payload and delegates blockchain-specific operations to the Fabric provider.

The intended transaction lifecycle includes payload preparation, chain code invocation, transaction submission, and retrieval of the required response. Record retrieval and verification support subsequent audit checks.

Unlike the mock provider, which simulates record operations within the application, the live provider communicates with the configured Fabric network. The deployed chain code name, channel, transaction functions, and network configuration should be documented from the actual environment. Transaction outputs and successful ledger queries should be retained as evidence.

5.3.3 Reassessment and Audit History

Welfare circumstances may change, requiring reassessment. The audit model supports references between earlier and subsequent records so that changes can be traced without silently removing historical context.

The exact record-versioning and status-transition behavior must be verified against the deployed chain code and integration tests. The thesis should describe the behavior demonstrated by the implementation rather than assuming every intended transition is supported.

5.4 Verification and Software Testing

Testing examined the backend services, eligibility assessment, audit operations, and blockchain-related functionality. Each testing level addresses different risks and should be reported separately.

5.4.1 Functional Testing

Functional tests evaluate expected behavior for supported inputs and workflows, including valid application data, invalid or incomplete values, eligibility boundaries, officer decisions, audit-record creation, retrieval, and reassessment.

The six-dimensional eligibility model should be tested using different input combinations and values near configured thresholds to check consistent application of the rules.

5.4.2 Blockchain Audit Testing

Blockchain audit tests examine record creation and retrieval, application-to-audit associations, and hash verification. Mock-provider tests verify audit logic in a controlled environment but cannot replace testing against the live Fabric network.

Live integration testing should establish that the backend communicates with Fabric, submits the intended transactions, and retrieves the expected ledger state. Reassessment and audit-history operations should also be tested where supported by the deployed chain code.

5.4.3 Integrity Verification Testing

Integrity tests should cover both unchanged and deliberately modified records. An unchanged record should produce a matching hash, while a modified record should produce a mismatch.

These tests demonstrate hash-verification behaviour under the tested conditions; they do not establish confidentiality or protection against every possible tampering scenario.

5.4.4 Test Evidence and Reporting

The existing thesis draft reports 64 passing tests out of 64, including 17 blockchain-audit tests, and zero TypeScript compilation errors. These figures should be retained only if supported by the relevant test report and final source revision.

Because live Fabric integration was completed subsequently, the final testing record should distinguish the earlier automated suite from tests executed against the integrated Fabric environment.

Testing should use controlled or synthetic records unless appropriate approval has been obtained to use real personal data. Results should be documented sufficiently to support reproducibility.

5.5 Deployment and Operation

The component was developed in a controlled environment and integrated with a configured Hyperledger Fabric network. Its operation depends on the backend services, provider configuration, network availability, and deployed chain code.

For an audit operation, the application prepares the payload and calculates its integrity hash before the configured provider handles blockchain interaction. Retrieval and verification operations allow the application to inspect the associated audit information.

Operational readiness also requires secure network configuration, appropriate credentials, access controls, logging, error handling, and recovery procedures before real welfare applications are processed.

The component should therefore be presented as a research prototype with completed Fabric integration, without implying that the entire SmartGrama platform has been approved or deployed for production government use.

6 RESULTS AND VALIDATION

6.1 Research Results

This chapter presents the results of implementing and evaluating the SmartGrama welfare eligibility and blockchain audit component. The evaluation focuses on configurable eligibility assessment, privacy-aware audit recording, officer decision-making, and integrity verification.

The evidence includes backend implementation, automated testing, hash verification, and the completed Hyperledger Fabric integration. Results from mock-provider testing are distinguished from those obtained through the live network.

6.1.1 Functional Assessment Results

The welfare eligibility service implements a configurable assessment model covering six dimensions: education, health and special needs, economic level and income, assets and ownership, housing conditions, and family demography. [4], [5]

The assessment generates a prototype score and category based on configured parameters. Testing should examine expected outputs and the handling of invalid, incomplete, and boundary-value inputs. The result remains advisory, with the authorized officer responsible for the final decision. [7]

The references cited in this chapter provide background and technical standards; implementation outcomes remain supported by the project test reports, transaction/query outputs, and integrity-verification evidence rather than by external literature.

The existing thesis records 64 passing automated tests out of 64, including 17 blockchain-audit tests. These figures apply to the documented test run and should not be treated as the final live Fabric test count unless confirmed by the relevant report.

6.1.2 Identity and Application Context

The component associate’s welfare applications with identity references and application records, connecting assessment and audit activities without reproducing complete personal records in every audit operation. [1], [2]

These results do not prove biometric accuracy unless the face-matching mechanism has been evaluated. Similarly, handling NIC or KYC information does not establish cryptographic authentication against an authoritative identity source. Claims must therefore reflect the identity functionality actually implemented and tested. [10]

6.2 Product and Technical Results

6.2.1 Backend and Audit Functionality

The backend provides services for eligibility assessment, audit-payload preparation, record management, and hash verification. The provider interface separates audit logic from blockchain-specific operations, enabling development and testing with a mock provider before integration with Hyperledger Fabric. [3]

This modular structure supports maintainability by isolating blockchain communication from the main eligibility logic.

6.2.2 Live Hyperledger Fabric Integration Results

The live Hyperledger Fabric integration has been completed, connecting the component to the configured Fabric environment. Its validation focuses on transaction submission, record retrieval, and integrity verification. [3]

A successful transaction demonstrates that the tested write operation was accepted through the network. A subsequent query can demonstrate record retrieval, while hash verification checks whether the retrieved data match the expected integrity value. [3]

Mock-provider tests demonstrate behaviour in a simulated environment, whereas live Fabric tests demonstrate behaviour against the configured network. These forms of evidence support different conclusions.

6.2.3 Hash-Verification Results

The component uses SHA-256 to calculate an integrity digest for audit payloads. Consistent payload preparation helps ensure that equivalent data produce the same digest when represented in the same canonical form. [9]

An unchanged payload should produce a matching digest, while a modified payload should produce a different digest and be identified by the verification mechanism. The

existing automated blockchain-audit tests provide evidence for the earlier implementation. For the final evaluation, verification should also be tested against records retrieved from Fabric.

Hash matching confirms consistency with the expected digest; it does not establish data accuracy, confidentiality, or eligibility-rule correctness.

6.2.4 Privacy-Oriented Payload Design

The component separates detailed welfare information from the smaller set of metadata needed for auditing. Potential fields include application and record references, DID reference, scheme and rule version, assessment outcome, record status, timestamps, and integrity hash. [1], [2]

The final payload must be checked against the actual implementation. Before claiming that no personal identifiable information is stored on-chain, the submitted transaction data and deployed chain code data model must be inspected.

6.3 User or Stakeholder Validation

Technical testing evaluates whether functions behave as expected, while stakeholder validation examines usability, clarity, administrative suitability, and perceived usefulness.

No formal stakeholder study should be conducted unless it was conducted and documented. If the evaluation was limited to software tests using controlled records, the findings represent technical validation rather than user acceptance.

Future evaluation could involve relevant officers or intended users reviewing the clarity of eligibility explanations, officer workflow, audit history, and verification outcomes, subject to appropriate research procedures and ethical safeguards.

6.4 Integration Results

Integration covers both the component's backend services and its connection to the live blockchain network. The eligibility result feeds into the officer decision workflow and audit preparation process, while the provider interface connects the application to Fabric.

The completed Fabric integration is a significant technical milestone, but it does not establish that the entire SmartGrama platform is production ready. Further validation may be required for policy rules, privacy controls, operational security, and integration with other group components.

6.5 Summary of Findings

The results show that the component has progressed from design to implementation and live blockchain integration. Its backend supports configurable welfare assessment, officer review, audit-payload preparation, and SHA-256-based integrity checking. The earlier implementation also has a documented automated test suite, while the live Fabric integration has subsequently been completed.

Three principal findings emerge. First, the component connects eligibility assessment with an explicit officer decision and associated audit information. Second, the provider abstraction separates application-level audit logic from blockchain operations. Third, the completed Fabric integration provides a technical connection for audit operations against the configured network.

These findings remain limited to the evidence available. The prototype assessment is not established as the official Aswesuma algorithm, and hash verification does not guarantee confidentiality or accuracy. Production readiness would require further evidence of security, operational reliability, and integration across the wider platform.

7 DISCUSSION

7.1 Interpretation of Findings

This chapter interprets the results presented in Chapter 6 in relation to the research problem and objectives. The component addresses the need for structured welfare eligibility assessment, human-supervised decision-making, privacy-conscious audit data, and integrity checking. The completed Hyperledger Fabric integration is considered alongside the earlier mock-provider testing.

The findings indicate that the component brings together several technical responsibilities that are often treated separately: calculating a prototype eligibility result, presenting that result for officer review, preparing audit metadata, and interacting with a permissioned blockchain. The contribution lies in the implementation and integration of these responsibilities within the SmartGrama component architecture.

The results should not, however, be interpreted as proof that the component reproduces official welfare policy or that the complete platform is production ready. There are different claims requiring different forms of evidence.

7.1.1 Interpretation of the Eligibility Workflow

The six-dimensional eligibility model provides a structured mechanism for processing application information. Configurable rules make it possible to adjust prototype parameters without redesigning the entire service, while the generated category and explanation provide information that can support review.

This design is relevant to the research problem because consistent execution of defined rules can reduce differences caused by inconsistent manual calculations. Nevertheless, consistency in software execution is not the same as policy correctness. The prototype's rules must be validated against authoritative requirements before the resulting classifications can be treated as suitable for actual welfare decisions.

The separation of the calculated result from the final officer decision is also important. It preserves a distinction between technical assessment and administrative authority. An automated result can inform a decision, but it should not be presented as an official grant or rejection unless the relevant policy and governance arrangements authorize that use.

7.1.2 Interpretation of Blockchain Integration

The completed live Fabric integration establishes that the component's blockchain interaction has progressed beyond a mock-only architecture. The provider abstraction supports a separation between business logic and network communication, while the

live provider enables the application to interact with the configured Fabric environment.

This is significant because mock-provider tests and live-network integration tests answer different questions. A mock provider can verify the behaviour of application logic in a controlled environment. A live integration test can additionally examine whether the application communicates with the configured network and whether the intended transaction or query operations behave as expected.

The conclusion drawn from live integration must still be tied to the operations tested. A successful writing operation does not automatically establish that every retrieval, reassessment, access-control, or failure-recovery path has been validated. The scope of each claim should therefore match the corresponding evidence.

7.1.3 Interpretation of Audit Integrity

The use of SHA-256 provides a mechanism for checking whether an audit payload matches its expected digest. When the payload is prepared deterministically, the verification process can compare the calculated digest with the value associated with the record.

This contributes to auditability because it can help identify changes to data that should remain consistent with the original record. However, a hash does not establish that the original information was correct or that the assessment was fair. Nor does it conceal the contents of data that may be publicly or improperly exposed.

The audit mechanism is consequently best understood as an integrity-control measure within a wider governance and security design.

7.2 Comparison with Literature, Baselines, and Existing Solutions

The component draws on established concepts from decentralized identity, welfare targeting, permissioned blockchain applications, and cryptographic integrity checking. The literature provides a conceptual basis for these design decisions, while the component's own test evidence determines what can be concluded about the implementation.

7.2.1 Decentralized Identity

The W3C Decentralized Identifiers specification describes a standard model for decentralized identifiers and associated mechanisms [1]. The W3C Verifiable Credentials Data Model provides a framework for representing claims that can be issued and verified under defined procedures [2].

These approaches are relevant because the welfare workflow requires a reliable relationship between an applicant's identity context and the application being assessed.

A decentralized identifier can provide a reference for that relationship, but the identifier alone does not prove a person's identity.

Identity assurance depends on the procedures used to establish the identity, validate evidence, issue credentials, and determine whether the presented information should be trusted. The component should therefore be assessed according to the identity-related mechanisms actually implemented, rather than assuming that using a DID reference automatically establishes full decentralized identity compliance.

7.2.2 Welfare Targeting and Eligibility Assessment

Research on proxy means testing and welfare targeting in Sri Lanka provides context for the challenges involved in assessing households and allocating benefits [3], [4]. These studies are relevant to the domain because welfare assessments may depend on several socioeconomic factors and carefully defined policy criteria.

The component applies a configurable prototype model across six welfare dimensions. This provides a way to study how structured input processing and rule-based calculations can support an application workflow.

The comparison must remain qualified: the prototype does not claim to reproduce the official production Aswesuma algorithm. Its scores and categories are research outputs based on the implemented configuration. Their policy validity depends on authoritative rules and domain-expert validation.

7.2.3 Permissioned Blockchain and Smart Contracts

Hyperledger Fabric supports permissioned distributed-ledger applications and smart contracts for application-specific business logic [5]. Such a platform can be relevant where authorized participants need to share and inspect a record of transactions or administrative actions.

The component's completed Fabric integration applies this approach to welfare audit operations. Unlike mock-provider testing, live integration makes it possible to test the application's communication with the configured ledger environment.

The strength of the comparison depends on the recorded integration evidence. A claim of successful submission, retrieval, or reassessment should be supported by the corresponding transaction output or test report. The mere presence of a Fabric provider is not enough to establish every operational property of the network.

7.2.4 Cryptographic Integrity and Privacy

The NIST Secure Hash Standard defines cryptographic hash functions, including SHA-256 [6]. The component uses hashing to support the verification of audit-payload integrity.

This is consistent with the purpose of a digest-based integrity mechanism, but it should not be confused with encryption or access control. A hash does not hide personal information, and an integrity check does not establish the truthfulness of the original data.

The privacy-oriented design therefore requires separate controls to minimize the information recorded on-chain and protect detailed personal information. These controls must be verified against the final payload and storage implementation.

The comparison shows that the component combines established technical principles in a welfare application context. Its contribution should be assessed through implementation quality, integration evidence, and the limits of its evaluation rather than through the use of technologies alone.

7.3 Research Contribution

The research contribution is the design and evaluation of a component that combines configurable welfare eligibility assessment, human-supervised decision-making, privacy-oriented audit-payload preparation, and blockchain-based record management.

First, the component structures the eligibility workflow around six dimensions and configurable parameters. This provides a basis for testing how a prototype assessment model can process welfare application information consistently and produce an interpretable result.

Second, the component distinguishes the calculated eligibility outcome from the final officer decision. This is important in a welfare context because a technical score should not automatically be treated as the final administrative determination.

Third, the component incorporates a minimized audit-payload design and SHA-256 integrity checking. This provides a mechanism for associating the application workflow with audit evidence while reducing the need to reproduce the complete personal record in the blockchain layer.

Fourth, the completed live Fabric integration demonstrates that the component's blockchain interaction has progressed beyond the earlier mock-provider environment. It provides a practical basis for evaluating audit operations against the configured permissioned network.

These contributions are primarily architectural and implementation-oriented. The current evidence does not establish a novel cryptographic algorithm, a formally validated welfare-targeting model, or the production readiness of the full SmartGrama platform.

7.4 Product Contribution

The product contribution concerns the role of the component within the SmartGrama platform. It provides backend functionality for eligibility assessment, officer review, audit-payload preparation, and blockchain-related operations.

The modular provider design separates blockchain interaction from the core eligibility service. This makes it easier to test application logic independently and to manage the network-specific responsibilities in a dedicated implementation.

The completed Fabric integration is an important product milestone because the component can interact with the configured live blockchain environment rather than relying only on simulated record storage. The extent of the resulting functionality must nevertheless be described according to the transaction, query, and verification operations actually validated.

The product also preserves a distinction between assessment and decision-making. This is relevant to administrative accountability because it supports review of the assessment result before the final decision is recorded.

The component should therefore be viewed as an integrated research prototype. Its wider operational value depends on reliable integration with the other SmartGrama components, correct policy configuration, privacy controls, security validation, and suitable operational procedures.

7.5 Limitations and Threats to Validity

Several limitations affect the interpretation and generalizability of the results.

7.5.1 Prototype Eligibility Rules

The eligibility model uses configurable research parameters and is not claimed to reproduce the official production Aswesuma PMT algorithm. A correctly executed calculation can still produce an unsuitable result if the underlying parameters do not match authoritative policy.

Future work should obtain appropriate policy specifications, document the source and version of each rule, and validate the implementation with domain experts. The thesis should not claim policy accuracy without such evidence.

7.5.2 Identity Verification

The strength of identity assurance depends on the mechanisms actually implemented. An identity reference or verification interface does not, on its own, establish accurate facial matching, authentic NIC verification, or resistance to identity fraud.

Where biometric or document-verification algorithms are implemented, their performance should be evaluated using suitable test data and clearly defined metrics. Without such evidence, the thesis should limit its conclusions to the demonstrated workflow functionality.

7.5.3 Scope of Live Fabric Testing

The live Fabric integration is completed, but the scope of the conclusions depends on the tests performed against the configured network. Successful integration does not automatically establish that every transaction type, error condition, security policy, or recovery scenario has been tested.

The final evaluation should document which write, query, hash-verification, and reassessment operations were exercised. It should also distinguish these results from the earlier mock-provider tests.

7.5.4 Privacy and Security

The use of SHA-256 does not provide confidentiality. Sensitive data can still be exposed if it is placed in an accessible payload or logged in an inappropriate location.

The privacy claims depend on the actual transaction payload, storage configuration, access controls, and treatment of identity references. The final implementation should be inspected to verify that unnecessary personal and biometric information is not written to the ledger.

7.5.5 Testing and Generalizability

Automated tests demonstrate behaviour under the tested conditions. They do not establish correctness for every input or operational scenario.

The existing thesis reports a 64-test suite, but the final thesis should associate each numerical result with the appropriate source revision and test report. Live Fabric test results should be reported separately if they were conducted after that suite was recorded.

If no formal stakeholder study was conducted, usability and administrative acceptance remain unvalidated. Similarly, testing with controlled or synthetic records does not establish performance with real citizen data or under production workloads.

7.6 Implications

7.6.1 Implications for System Design

The component illustrates the value of separating eligibility assessment, officer decision-making, audit-payload preparation, and blockchain interaction. This separation supports independent testing and can reduce the coupling between policy logic and network-specific code.

The provider abstraction also makes it possible to use mock-provider tests during development while retaining a separate live Fabric implementation for integration. This is useful because application-level logic can be tested without requiring every test to depend on network availability.

7.6.2 Implications for Welfare Administration

A structured assessment result and associated audit information may help authorized personnel examine how an application progressed through the system. Linking records to applications and maintaining appropriate reassessment history can support review of changes over time.

However, an auditable record is useful only when the information is meaningful, accessible to authorized personnel, and governed by clear procedures. The system should define how records are reviewed, how reassessments are represented, and how disputes or corrections are handled.

The component should support administrative accountability rather than replace the authority of the responsible decision-maker.

7.6.3 Implications for Privacy and Responsible Use

Welfare applications may contain sensitive information about identity, health, income, household circumstances, and assets. The architecture should minimize the data placed on the blockchain and retain detailed personal records in appropriately protected storage.

Privacy protection should be treated as a system-level responsibility. It requires data minimization, access controls, suitable retention procedures, secure configuration, and careful management of references that might allow records to be linked.

The potential consequences of incorrect eligibility assessment or identity verification also require attention. Incorrect rejection can affect an eligible applicant, while incorrect acceptance may affect the fair allocation of benefits. Clear explanations, appropriate officer review, and suitable dispute-handling processes are therefore important.

7.6.4 Implications for Future Development

Future development should prioritize validating the eligibility parameters against approved policy, evaluating the actual identity-verification mechanisms, and expanding the test coverage of the completed Fabric integration.

Further work should also assess security, performance, reliability, privacy, and the component's interactions with the wider SmartGrama platform. A formal stakeholder evaluation could provide evidence about whether the workflow and audit information are understandable and useful to intended users.

In conclusion, the component combines configurable welfare assessment, officer-supervised decisions, integrity verification, and completed live Hyperledger Fabric integration. The results provide a foundation for further validation, while the limitations clarify why successful technical integration should not be equated with official policy validation or full production readiness.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value Proposition and Target Users

8.1.1 Introduction

The commercialisation and sustainability of the SmartGrama Natural Language Processing (NLP) component depend on its ability to provide useful, accessible, and maintainable information services for people interacting with welfare and microfinance systems. The component was developed to support users in obtaining answers to questions related to welfare services and microfinance through natural-language interaction in English and Sinhala. By combining a question-answering approach with retrieval-augmented generation (RAG), the prototype aims to make relevant information easier to access without requiring users to navigate complex documentation or understand technical terminology.

The potential value of the component is not limited to commercial profit. In the context of a public-oriented platform such as SmartGrama, value may also be demonstrated through improved access to information, reduced repetitive enquiries, more consistent responses, and support for users who prefer communicating in Sinhala. However, these benefits represent intended outcomes and opportunities for future validation rather than confirmed operational impacts. The current prototype and its documented evaluation provide a foundation for further development, but real-world adoption would require additional technical, organisational, and user-centred validation.

8.1.2 Value Proposition

The main value proposition of the NLP component is to provide a conversational interface through which users can seek information about welfare and microfinance services using natural language. Traditional information-access approaches may require users to search websites, read lengthy documents, contact an officer, or understand formal administrative terminology. A conversational system offers an alternative interaction method by allowing users to submit questions in a familiar language.

The principal value propositions are described below.

Improved information accessibility: The chatbot is intended to make welfare-related information easier to access by allowing users to ask questions directly rather than manually searching multiple information sources.

Bilingual interaction: Support for English and Sinhala can make the system more accessible to users with different language preferences. The usefulness of this feature depends on the quality of language handling, the coverage of the knowledge base, and the correctness of the generated answers.

Retrieval-supported answers: The RAG approach retrieves relevant information from an indexed knowledge source before generating an answer. This design is intended to make responses more closely connected to available reference material than generation based only on a language model's general knowledge. Nevertheless, retrieval does not guarantee correctness, and unsupported or inaccurate responses remain possible.

Potential reduction of repetitive enquiries: If adopted by a welfare service organisation, the chatbot could help answer common questions and allow staff to concentrate on enquiries that require human judgement. The extent of any workload reduction would need to be measured through a real-world pilot.

Flexible technical architecture: The use of a retrieval layer, FAISS vector search, sentence-transformer embeddings, and a locally operated language model through Ollama provides a modular technical foundation. Individual elements can potentially be updated as the knowledge base, language models, or operational requirements change. Such changes would still require compatibility testing and evaluation.

Integration potential: As part of the wider SmartGrama platform, the NLP component could serve as an information-access layer alongside other system components. Its role is to assist users in understanding services and available information; it should not independently make official welfare eligibility decisions unless such a role is separately authorized, designed, and validated.

8.1.3 Target Users and Beneficiaries

The potential target users include individuals seeking welfare or microfinance information, public-facing service officers, administrators responsible for maintaining information, and organizations operating digital public-service platforms.

Table 11 : Target Users

Target group	Potential value	Requirements before adoption
Welfare applicants and beneficiaries	Easier access to information about welfare services and related procedures	Clear language, accessible interaction, reliable answers, and appropriate privacy safeguards
Sinhala-speaking users	Access to information through Sinhala-language questions	Evaluation of Sinhala comprehension, spelling variations, and answer quality
Welfare and administrative officers	Potential assistance with repetitive information enquiries	Verified knowledge sources, escalation to officers, and clear limits on automated responses

Platform administrators	A centralized mechanism for providing information through a chatbot	Content-management procedures, monitoring, access controls, and maintenance resources
Public-service organizations	Potential improvement in digital information delivery	Organizational approval, integration planning, security review, and pilot evaluation

These groups represent potential users and beneficiaries. Their inclusion in this analysis does not imply that the component has already been deployed to them or that formal consultations have been conducted.

8.1.4 Differentiation from a Conventional Information Website

A conventional website generally presents information through menus, search fields, and predefined pages. The SmartGrama NLP component instead aims to interpret a user's natural-language question, retrieve relevant information, and generate a conversational response. This approach may be useful when users do not know the exact title of a document or the administrative term associated with their question.

The distinction is one of interaction method rather than a claim that a chatbot is always more effective than a website. A website can provide stable, directly verifiable information, while a conversational interface can simplify information discovery. A future operational version could therefore complement existing websites and human assistance rather than replace them entirely.

8.2 Market and Competitive Position

8.2.1 Potential Application Environment

The initial application environment for the component is the digital delivery of welfare and microfinance information. Organizations that provide these services may need to communicate eligibility requirements, application procedures, required documents, deadlines, and other service information to diverse groups of users.

The potential application environment includes public-sector digital services, community-oriented information platforms, non-governmental organizations, and institutions providing financial or welfare assistance. These are prospective application areas, not confirmed by customers or partners of the research project.

The need for accessible digital information provides a rationale for exploring conversational interfaces. However, this study has not established a quantified market size, conducted a formal competitor survey, or demonstrated demand through a

commercial pilot. Consequently, the commercial opportunity should be treated as a hypothesis requiring further investigation.

The proposed differentiating characteristics are the focus on welfare and microfinance information, bilingual interaction, retrieval-supported response generation, and the possibility of local language-model operation. These characteristics may be relevant where organizations require greater control over the information source and deployment environment.

Using a locally operated language model may offer greater control over where inference takes place and may reduce reliance on an external hosted model API. However, local operation does not automatically guarantee privacy, low cost, high availability, or strong performance. The hosting organization would still need to manage infrastructure, access controls, software updates, logging, and security.

The component's competitive position should therefore be assessed through practical criteria such as answer correctness, retrieval relevance, language coverage, response time, operating cost, ease of maintenance, and user accessibility. These measures would provide a stronger basis for comparison than general claims of superiority.

8.3 Business or Sustainability Model

8.3.1 Proposed Sustainability Approach

The NLP component is currently a research prototype rather than an established commercial product. No completed commercial launch, revenue stream, paid customer agreement, or formal deployment partnership is claimed in this thesis. Its sustainability should therefore be considered in terms of the resources and organizational arrangements required to maintain and responsibly operate it.

A suitable future model would combine technical maintenance with controlled content management. The knowledge base would need to remain aligned with current welfare and microfinance information. The embedding model, vector index, language model, retrieval configuration, and response-generation logic would also need periodic evaluation as source documents and operational requirements change.

For a public-service application, sustainability could be supported through an institutionally maintained service rather than relying exclusively on direct payment by individual users. Potential arrangements include:

- **Institutional operation:** A public-service organization or authorized service provider operates the chatbot as part of its existing digital information services.

- **Shared platform service:** The NLP component is maintained as a reusable service within the wider SmartGrama platform, subject to agreed technical and governance responsibilities.
- **Grant or research support:** Early-stage development, accessibility improvements, and pilot evaluation may be supported through research funding or relevant digital-development programmers.
- **Managed technical support:** An authorized technical team provides maintenance, knowledge-based updates, monitoring, and integration support under a future service arrangement.

These are proposed options, not existing funding or business arrangements.

8.3.3 Maintenance and Knowledge Management

The quality of a retrieval-augmented question-answering system depends substantially on the information available to it. Outdated, incomplete, ambiguous, or conflicting documents may result in poor retrieval and misleading responses even when the software functions as intended.

A sustainable operational process should include the following activities:

1. Identify authorized and trustworthy information sources.
2. Review documents before adding them to the knowledge base.
3. Record source ownership and, where possible, publication or review dates.
4. Process approved documents and regenerate embeddings when required.
5. Rebuild or update the FAISS index following material knowledge-base changes.
6. Test representative English and Sinhala questions after each significant update.
7. Review unsupported answers and document corrective actions.
8. Provide a route for users to seek assistance from an authorized officer when the chatbot cannot answer reliably.

These activities would help keep the system aligned with current information. The exact update frequency should be determined by the responsible organization and the rate at which relevant policies and procedures change.

8.3.4 Measures for Long-Term Sustainability

A future pilot should assess sustainability using measurable indicators rather than relying only on the availability of the software. Suitable indicators include:

- Proportion of questions answered correctly against a reviewed reference set.
- Retrieval relevance for representative English and Sinhala queries.
- Frequency of unsupported or misleading responses.
- Response time under expected operating conditions.
- Cost per query or per defined period of operation.
- Frequency of knowledge-base updates and unresolved content issues.
- Availability of the service during agreed operating periods.
- Number and type of enquiries successfully resolved without unnecessary escalation, where this can be measured ethically and reliably.

The existing 217-question evaluation provides a basis for structured assessment, but its documented findings should not be interpreted as proof of real-world effectiveness or financial sustainability. Further controlled testing and an operational pilot would be needed to establish those outcomes.

8.4 Scalability, Intellectual Property and Compliance

8.4.1 Technical Scalability

The modular design of the NLP component provides a potential foundation for future expansion. The retrieval process, vector index, language-model interface, and application integration can be considered as separate technical responsibilities. This separation may make it easier to maintain individual components without redesigning the complete system.

Nevertheless, scalability is not established solely by modular architecture. Increasing the number of users or documents may increase processing time, memory consumption, storage requirements, and concurrent-request demands. Local language-model inference may become a particular constraint if the available computing resources cannot support the expected workload.

Before wider deployment, the system should undergo load testing with realistic query volumes and document collections. The results should be used to determine suitable hardware, concurrency limits, catching requirements, monitoring arrangements, and any need for distributed deployment. No claim of large-scale or production-level performance is made without such evidence.

8.4.2 Intellectual Property and Licensing

The intellectual-property position of the component should be clarified before commercial distribution or institutional adoption. The software developed specifically for the research project may be subject to university rules, group-project agreements, and the contributions of individual team members. Ownership and permitted reuse should therefore be confirmed with the relevant academic and project authorities.

The component also relies on third-party technologies and models, including sentence-transformer embeddings, FAISS vector search, and Ollama-based local model operation. Each dependency and selected model must be checked against its applicable licence and usage conditions. Open-source availability does not necessarily mean that every use, modification, or commercial distribution is unrestricted.

A future release should include a software bill of materials or equivalent dependency inventory, licence records, attribution requirements, and documentation of modifications. The use of any model for a public or commercial service should be assessed against that model's specific licence and deployment conditions.

8.4.3 Privacy and Data Protection

Because the intended application concerns welfare and microfinance information, privacy should be considered throughout the design and operation of the service. Questions submitted by users may contain names, identification details, financial circumstances, or other sensitive information. Such details should not be requested or retained unless they are necessary for an explicitly defined and authorized purpose.

A future deployment should apply data minimization, purpose limitation, appropriate access controls, secure configuration, and retention rules. Logs used for debugging or evaluation should be reviewed to avoid unnecessary storage of personal information. If external model services are introduced, the organization should assess where submitted data is processed and whether the arrangement meets its privacy requirements.

Local inference may reduce the need to send questions to an external model provider, depending on the final architecture. It does not remove the need for security controls on the application, host machine, knowledge base, logs, and administrative interfaces.

The deployment organization should assess applicable Sri Lankan data-protection and other legal requirements with qualified advisers before operational use. This thesis does not claim that the prototype has received legal certification or that compliance has been independently audited.

8.4.4 Responsible Use and Operational Governance

A welfare information chatbot should support access to information without misleading users about the status of their applications or their entitlement to assistance.

The answers generated must not be presented as official decisions unless they have been authorized through the relevant administrative process.

The system should communicate its limitations and provide an escalation path when information is missing, ambiguous, or outside its intended scope. In particular, a generated response should not replace an authorized officer's decision on an individual welfare application.

The documented evaluation identified unsupported responses concerning registration information as a limitation requiring further improvement. This issue is commercially and operationally important because users may rely on incorrect procedural guidance when applying for assistance. Improvements should include stronger source coverage, response validation, clearer refusal or escalation behavior, and repeated testing of questions that previously produced unsupported answers.

8.4.5 Organizational Adoption and Future Deployment

Successful adoption would require more than software completion. A potential host organization would need to identify an accountable service owner, approve the information sources, define maintenance responsibilities, and establish procedures for correcting errors and handling user complaints.

A staged adoption approach is recommended:

1. Technical readiness review: Verify dependencies, security settings, deployment requirements, and reproducibility of the evaluation.
2. Knowledge-base review: Confirm the accuracy, currency, and authority of the information sources.
3. Controlled pilot: Evaluate the system with approved test questions and, only after appropriate ethical and privacy review, a carefully designed user study.
4. Operational monitoring: Record suitable non-sensitive quality and reliability indicators and investigate reported errors.
5. Expansion decision: Consider broader use only after the pilot demonstrates acceptable performance and the responsible organization approves the operating model.

This staged approach reduces the risk of presenting a research prototype as a fully validated public-service solution.

8.5 Contribution to the Sustainable Development Goals

The potential social value of the SmartGrama NLP component can be discussed in relation to the United Nations Sustainable Development Goals (SDGs). These

connections describe intended areas of contribution; they do not establish that the prototype has already produced measurable development outcomes.

8.6 Chapter Summary

This chapter examined the commercialization and sustainability potential of the SmartGrama NLP component. The proposed value lies in conversational access to welfare and microfinance information, English and Sinhala interaction, retrieval-supported response generation, and the potential integration of the chatbot into a wider digital-service platform.

The analysis identified possible user groups, alternative information-delivery approaches, sustainability models, cost categories, and requirements for scalability, intellectual-property management, privacy, and responsible operation. It also discussed potential relationships with SDGs 1, 10, 16, and 9. These opportunities remain prospective because the component has not been established as a commercially launched or production-deployed service.

Long-term sustainability will depend on reliable information sources, continuous evaluation, maintainable infrastructure, clear organizational responsibility, and appropriate safeguards for users. The documented 217-question evaluation provides a basis for further technical assessment, while limitations such as unsupported registration-information responses highlight the need for continued improvement. A controlled pilot and a structured operational feasibility assessment are recommended before wider adoption.

Overall, the component demonstrates a research-oriented approach to improving access to welfare information through NLP. Its future value will depend not only on technical functionality but also on demonstrated answer quality, responsible governance, affordability, and the ability to meet the needs of its intended users.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Introduction

This chapter concludes the research and development of the SmartGrama Natural Language Processing (NLP) component, an English–Sinhala AI assistant designed to support access to welfare and microfinance information. The chapter revisits the research problem and objectives, summarizes the developed solution and evaluation, discusses the main contributions, and identifies the limitations that remain. It concludes with recommendations for future development and a final statement on the significance of the research.

The component was developed within the broader SmartGrama platform, which aims to improve access to digital services related to welfare and microfinance. Its specific role is to allow users to submit natural-language questions and receive information-oriented responses based on an available domain knowledge base. The research combines semantic retrieval, multilingual processing, and large language model generation to investigate how conversational AI can support information access in this domain.

The findings demonstrate the technical feasibility of developing a domain-focused conversational prototype. However, technical functionality alone does not establish that all generated answers are correct, that the system is suitable for unrestricted public use, or that it has produced measurable improvements in welfare-service delivery. These questions require further evaluation and real-world validation.

9.2 Summary of the Research

9.2.1 Research Problem

Accessing welfare and microfinance information can require users to navigate multiple documents, understand formal terminology, or seek assistance from service personnel. Language differences and unfamiliar administrative procedures may create additional barriers for people seeking information about available services, required documentation, registration, repayment, and related processes.

Conventional information-access methods remain important, but they may not always provide the conversational interaction some users prefer. A natural-language assistant offers an alternative by allowing users to express a question in their own words and receive a response through a conversational interface.

However, applying generative AI to welfare-related information introduces its own challenges. A language model may generate plausible statements that are not supported by the available documents. Questions written in different languages may also produce different retrieval and response-quality outcomes. Therefore, the research needed to

consider not only the ability to generate responses but also retrieval relevance, factual support, language quality, completeness, and response time.

9.2.2 Research Aim

The main aim of the research was to design, develop, and evaluate an English–Sinhala NLP assistant that uses retrieval-augmented generation to support access to welfare and microfinance information within the SmartGrama platform.

The work focused on integrating natural-language interaction with a domain-specific knowledge base so that responses could be informed by retrieved information rather than relying exclusively on the general knowledge of a language model.

9.2.3 Research Objectives

The research addressed the following objectives:

1. To investigate the requirements and challenges associated with conversational access to welfare and microfinance information.
2. To design an NLP architecture that combines semantic retrieval and language-model-based response generation.
3. To develop a bilingual conversational prototype using suitable embedding, retrieval, translation, and language-model technologies.
4. To evaluate the developed component using a structured collection of English and Sinhala questions.
5. To identify technical limitations and recommend improvements for reliability, accessibility, maintainability, and future adoption.

These objectives connect the research problem to the technical implementation and evaluation described in the preceding chapters.

9.3 Summary of the Developed Solution

9.3.1 System Architecture

The developed prototype combines a React-based frontend with a Python Flask backend. The backend coordinates question processing, retrieval, and response generation. Sentence-transformer embeddings are used to represent text semantically, while FAISS supports similarity-based retrieval from the indexed knowledge base. The language-model component uses Ollama to operate a locally hosted model, and the implementation includes translation support for English–Sinhala interaction.

The architecture separates user interaction from the processing pipeline, allowing the interface, retrieval process, and response-generation components to be maintained and evaluated as related but distinct elements.

The principal processing stages are:

1. The user submits a question through the conversational interface.
2. The backend processes the input and applies the configured language-handling procedure.
3. The retrieval component searches the knowledge base for relevant information.
4. The retrieved context is provided to the language model as part of the response-generation process.
5. The generated response is returned to the user through the frontend.

The effectiveness of this process depends on the quality of the source documents, language processing, retrieval configuration, and response-generation behaviour. The presence of a retrieval stage improves the opportunity to ground responses in domain information but does not eliminate the possibility of unsupported or inaccurate answers.

9.3.3 Individual Research Contribution

The individual contribution addressed in this thesis is the SmartGrama NLP assistant rather than the complete SmartGrama platform. The research contribution includes the design and development of the NLP processing workflow, the integration of semantic retrieval and language-model response generation, bilingual question handling, and the structured evaluation of the resulting prototype.

The component is intended to provide informational assistance. It should not be interpreted as an independent authority for approving welfare applications, making official eligibility decisions, determining creditworthiness, or replacing authorized human judgement.

The component's value therefore lies in investigating a practical approach to making domain information accessible through conversational interaction, while identifying the quality and governance requirements that must be met before wider deployment.

9.4 Summary of Evaluation and Findings

9.4.1 Evaluation Approach

The component was evaluated using a structured collection of 217 question records covering relevant welfare and microfinance topics. The evaluation material includes fields for the language, category, question, expected answer, generated answer, retrieved context, and response time. This structure supports analysis of different aspects of system behaviour rather than treating successful response generation as the sole indicator of performance.

The evaluation considered dimensions including correctness, relevance, groundedness, completeness, retrieval behaviour, language quality, and response time. These dimensions are important because a response can be fluent and relevant in appearance while still containing unsupported information.

9.4.2 Technical Execution

The recorded evaluation workflow processed all 217 questions. The available evaluation notes report an overall average response time of approximately 4.77 seconds, with averages of approximately 4.14 seconds for English questions and 5.58 seconds for Sinhala questions.

These measurements provide an indication of processing performance under the conditions of the recorded evaluation. They should not be interpreted as guaranteed production response times because results may vary with hardware, model configuration, concurrent users, network conditions, and question complexity.

Successful processing of all evaluation questions also does not establish that every response was correct or complete. Pipeline completion and answer quality are separate evaluation outcomes.

9.4.3 Main Technical Findings

The research produced several important findings.

First, the combination of semantic retrieval and language-model generation provides a practical architecture for a domain-specific conversational assistant. The retrieval layer creates a mechanism for locating relevant information before a response is generated.

Second, the prototype supports a workflow intended to accommodate English and Sinhala questions. However, bilingual capability must be evaluated separately for each language because differences in wording, translation, spelling, and semantic representation may affect retrieval and response quality.

Third, response quality depends on the knowledge base as well as the language model. Missing, outdated, or insufficiently detailed source information can prevent the system from answering a question reliably. A retrieved passage may also be related to the question without containing enough evidence to support the complete answer.

Fourth, the evaluation identified unsupported responses concerning registration information. This illustrates an important limitation of generative systems: a response may include details that are not justified by the available source material. Such behavior is particularly significant in welfare-related contexts, where users may rely on procedural information when seeking assistance.

Finally, the recorded response-time measurements provide a useful baseline for future optimization. Further testing is needed to determine how processing time changes under larger workloads and different deployment configurations.

9.5 Research and Technical Contributions

9.5.1 Research Contribution

The research contributes an applied investigation of retrieval-augmented conversational AI for welfare and microfinance information. It considers how semantic search and language-model generation can be combined in a domain-focused assistant and how the resulting behaviour can be examined using a structured question set.

The evaluation approach also highlights the importance of assessing correctness, source support, completeness, language quality, and response time separately. This is particularly relevant to information services where a fluent response may still be misleading if it introduces unsupported procedural details.

9.5.2 Technical Contribution

The technical contribution is the development and integration of the NLP processing workflow into a usable conversational prototype. The implementation combines frontend interaction, backend orchestration, semantic embeddings, vector-based retrieval, translation support, and language-model generation.

The modular arrangement provides a basis for future changes to the retrieval strategy, knowledge base, or language model. Such changes should be accompanied by regression testing to establish whether they improve or degrade the quality of the responses.

9.5.3 Practical Contribution

The practical contribution is a prototype through which users can explore welfare and microfinance information using natural-language questions. Its English–Sinhala

interaction approach may be useful in environments where language preference affects access to digital information.

The practical significance remains conditional on answer reliability, accessibility, source quality, and organizational readiness. Further evaluation with representative users and approved information sources is necessary before the system can be considered suitable for routine public-service use.

9.6 Research Limitations

Despite the progress made, several limitations affect the interpretation and generalizability of the findings.

9.6.1 Knowledge-Base Coverage

The assistant is limited by the information contained in its knowledge base. Questions about missing, outdated, or insufficiently detailed material may produce incomplete answers or unsupported statements. A larger knowledge base alone would not necessarily solve this problem; the information must also be accurate, authoritative, current, and appropriately indexed.

9.6.2 Generated-Response Reliability

Retrieval-augmented generation does not guarantee factual correctness. The language model may misinterpret retrieved context, omit relevant details, combine unrelated information, or generate statements that are not supported by the retrieved passages. The identified registration-information responses demonstrate why answer validation and uncertainty handling remain important areas for improvement.

9.6.3 Bilingual Performance

The inclusion of English and Sinhala interaction does not establish equal performance across both languages. Translation, spelling variation, sentence structure, and the representation of Sinhala text may affect retrieval and response generation. Larger and more diverse language-specific evaluation sets are required to assess these issues comprehensively.

9.6.4 Evaluation Scope

The evaluation used a structured collection of 217 questions. Although this provides a basis for controlled assessment, it cannot represent every question, phrasing variation, user characteristic, or real-world situation. The evaluation should therefore be interpreted in the context of its question set and testing conditions.

The completion of the evaluation workflow should not be treated as an answer-accuracy score. Claims about accuracy, groundedness, or overall success require

explicit scoring rules, reproducible calculations, and a clearly documented evaluation report.

9.6.5 User and Organisational Validation

The available evidence does not establish a formal user study demonstrating satisfaction, improved access, reduced officer workload, or increased welfare-service uptake. These outcomes require a separately designed evaluation with appropriate participant selection, consent, privacy protections, and measurement procedures.

9.6.6 Production Readiness

The research prototype has not been established by the evidence presented here as a fully deployed, production-ready public-service system. Wider adoption would require further security testing, operational monitoring, accessibility evaluation, infrastructure planning, knowledge-governance procedures, and assessment of applicable legal requirements.

These limitations define the boundaries of the current findings and provide a clear basis for future work.

9.7 Future Directions

Future development should prioritise the reliability and responsible use of the assistant before expanding its functionality or user base.

9.7.1 Improving Knowledge-Base Quality

The first priority is to improve the coverage and quality of the source material. Future work should identify authoritative welfare and microfinance documents, review them for currency and consistency, and establish a process for updating the knowledge base.

The system should also distinguish between questions that can be answered from the available sources and questions for which sufficient evidence is missing. Where the information is unavailable, the preferred behaviour is to communicate that limitation rather than generate unsupported procedural details.

9.7.2 Improving Retrieval and Groundedness

Retrieval quality could be improved through systematic testing of text-chunk size, overlap, embedding-model selection, similarity thresholds, and ranking strategies. These changes should be compared against a documented baseline rather than introduced solely on the assumption that a more complex retrieval method will perform better.

A future evaluation should measure whether the retrieved passages contain the evidence required to answer each question. Response-generation evaluation should then determine whether the final answer accurately represents that evidence.

9.7.3 Strengthening Response Validation

The response-generation pipeline could incorporate additional checks to detect unsupported claims, incomplete answers, and contradictions with retrieved source material. The assistant should be able to decline to provide a definitive answer when the available evidence is insufficient.

Questions that previously produced unsupported registration information should be retained as regression tests. This would help determine whether future changes resolve known failures without introducing new problems in other categories.

9.7.4 Improving English–Sinhala Processing

Future work should expand language-specific testing and investigate common spelling variations, alternative phrasings, mixed-language queries, and differences in sentence structure. The evaluation should report English and Sinhala results separately as well as overall results.

Where translation is used, its influence on retrieval and answer quality should be examined. Direct multilingual retrieval and alternative embedding models could also be investigated through controlled comparisons.

9.7.5 Expanding the Evaluation Framework

The 217-question collection provides a starting point for a more comprehensive evaluation framework. Future work should define explicit scoring criteria, establish a consistent review procedure for expected answers, and calculate reproducible results for correctness, relevance, groundedness, completeness, and language quality.

Testing should also include questions that are ambiguous, outside the knowledge base, potentially misleading, or phrased in ways not represented in the original evaluation set. Performance should be measured under clearly documented hardware and configuration conditions.

9.7.6 Performance and Scalability Testing

Future performance testing should examine response times under concurrent requests, larger knowledge bases, and different model configurations. The results could guide decisions about caching, resource allocation, model optimization, and deployment architecture.

Any optimization should be evaluated against answer quality as well as processing time. A faster response is not an improvement if it substantially increases the risk of inaccurate or unsupported information.

9.7.7 Privacy, Security, and Responsible Operation

Further work should review how questions, generated responses, error logs, and knowledge-base documents are handled. Data minimization, access controls, retention policies, secure configuration, and appropriate monitoring should be incorporated into any deployment plan.

9.8 Final Conclusion

This research investigated the development of an English–Sinhala NLP assistant for welfare and microfinance information within the SmartGrama platform. The developed prototype combines a React frontend, Flask backend, semantic embeddings, FAISS-based retrieval, translation support, and locally operated language-model generation through Ollama. The implementation demonstrates a practical way to connect natural-language questions with a domain-specific information source through a retrieval-augmented workflow.

The documented evaluation processed 217 question records and provided a basis for examining system execution, response time, language-related behaviour, and answer quality. The reported average response times offer an initial performance reference, while unsupported responses concerning registration information reveal that reliability remains a central challenge. These findings reinforce the need to evaluate factual support and answer quality independently from successful processing.

The principal contribution of the research is an applied NLP prototype and an evaluation approach for conversational access to welfare and microfinance information. The component offers potential value through natural-language interaction and English–Sinhala support, but its effectiveness in real-world public-service settings has not yet been established by the available evidence.

Future progress should focus on authoritative knowledge sources, improved retrieval, stronger response validation, bilingual evaluation, privacy safeguards, scalability testing, and controlled user-centred validation. With these improvements, the component could provide a foundation for more accessible digital information services. Its eventual success will depend not only on technical capability but also on the trustworthiness of its responses, responsible operation, and demonstrated usefulness to its intended users.

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GLOSSARY

DID - Decentralized Identifier: an identifier used in decentralized identity systems.

KYC - Know Your Customer: identity/evidence verification used before processing the application.

PMT - Proxy Means Test: an approach that estimates household welfare using observable characteristics.

Chain code - Smart-contract code executed in a Hyperledger Fabric network.

Payload Hash - A cryptographic digest used to provide integrity evidence for audit data.

Rule Version - An identifier specifying the configuration of rules and thresholds used for an assessment.

Verifiable Credential - A structured digital credential that can express claims in a machine-verifiable form.

Human-in-the-loop - A design in which automated processing supports a human decision-maker.

APPENDICES

Appendix A Research instruments or extended results

A.1 Example eligibility input structure: applicationId, DID reference, scheme, ruleVersion, and the six assessment dimensions (education, health and special needs, economic level and income, assets and ownership, housing conditions, and family demography).

A.2 Example audit payload: recordId, applicationId, DID reference, scheme, ruleVersion, calculatedScore, category, eligibilityStatus, thresholdReference, recordStatus, previousRecordId when applicable, timestamp and payloadHash.

A.3 Reported automated testing: blockchain audit 17/17 passed; eligibility/core/integration and related workflow tests contributed to a reported total of 64/64 passed; TypeScript build reported zero compiler errors.

A.4 Recommended evidence for final submission: screenshots of the eligibility form, assessment result, officer review, blockchain record/transaction, verification result, test output and final integrated Fabric transaction where available.

Appendix B Academic integrity and AI use disclosure

Generative AI and automated tools were used as drafting and language-support tools during thesis preparation for activities such as organization, wording refinement and identifying areas requiring clarification. The candidate remains responsible for verifying technical descriptions, citations, implementation claims, test results, data, analyses and conclusions. The exact wording and evidence of AI use should be aligned with the module's current academic integrity policy before submission.